

Basic Power Electronics

Arnold Knott

DTU Electro, knott@dtu.dk

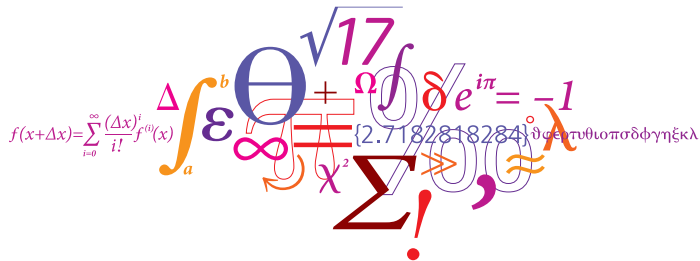


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Repetition:

- **Passive Components**

- Resistors
- Inductors
- Capacitors

- **Active Components - Power Semiconductors**

- Diodes
- MOSFETs
- Cooling
- Other Power Semiconductors

Components:

- **Nonisolated Converters**

- Buck Converter
- Boost Converter
- Buck-boost Converter

- **Parameters of Converters**

- Discontinuous Conduction Mode
- Synchronous Rectification
- Interleave

- **Isolated Converters**

- Isolation
- Flyback

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Main course literature: [1]

Part I

Repetition

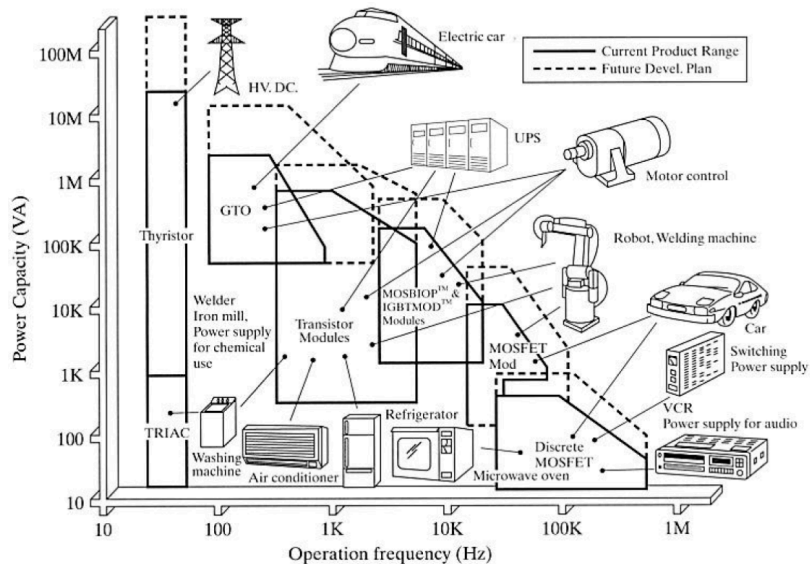
Section 1

Introduction

- **Introduction**
 - Applications
 - Properties of Power Supplies
 - Linear Solutions
 - Switch-Mode Solutions
- **Signals**

Introduction

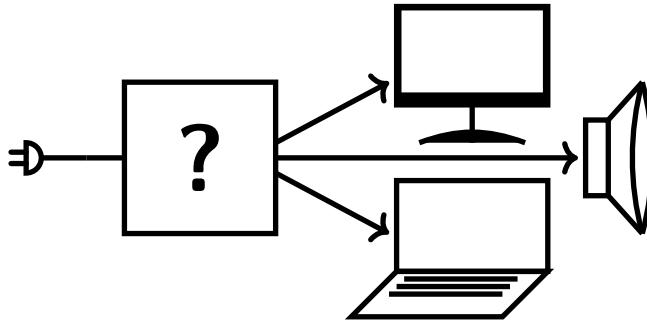
Applications



[2]

Exercise 1.1 Properties of power converters

What would you, as a user and/or as a coming engineer expect from power converters?



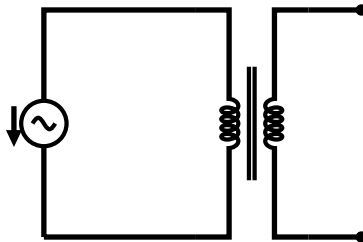
Introduction

Transformer

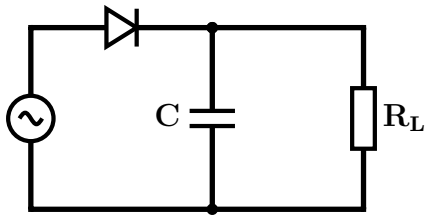
Transformer

US: 110 V_{RMS}, 60 Hz

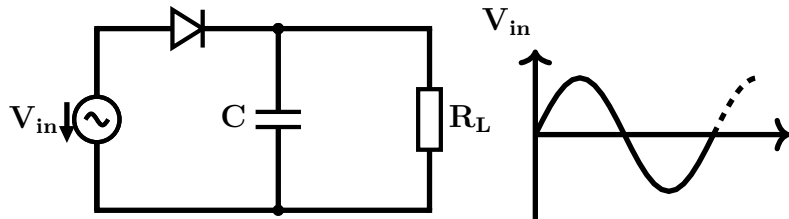
EU: 230 V_{RMS}, 50 Hz



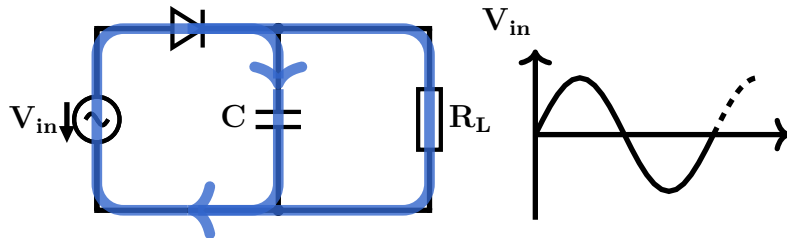
Halfwave rectifier



Halfwave rectifier

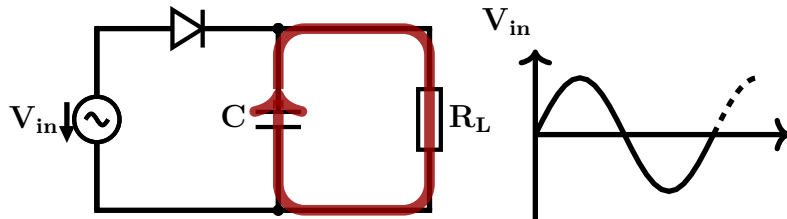


Halfwave rectifier



$V_{in} > V_C$:
 V_{in} is charging capacitor
and supplying load

Halfwave rectifier



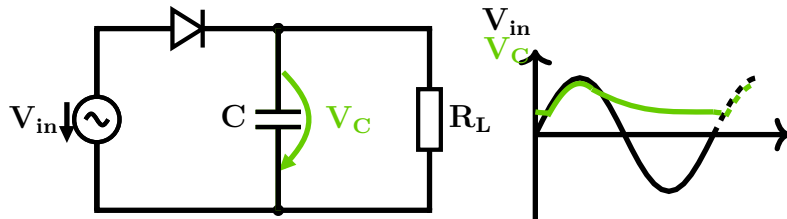
$V_{in} > V_C$:

V_{in} is charging capacitor
and supplying load

$V_{in} < V_C$:

Capacitor is supplying load

Halfwave rectifier



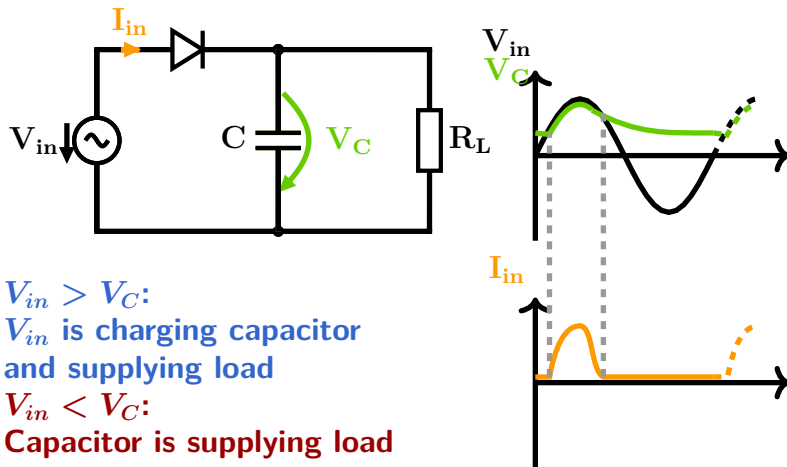
$V_{in} > V_C$:

V_{in} is charging capacitor
and supplying load

$V_{in} < V_C$:

Capacitor is supplying load

Halfwave rectifier



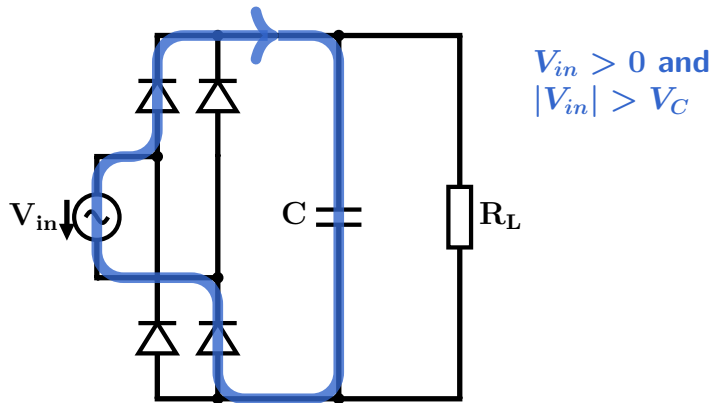
$V_{in} > V_C$:

V_{in} is charging capacitor
and supplying load

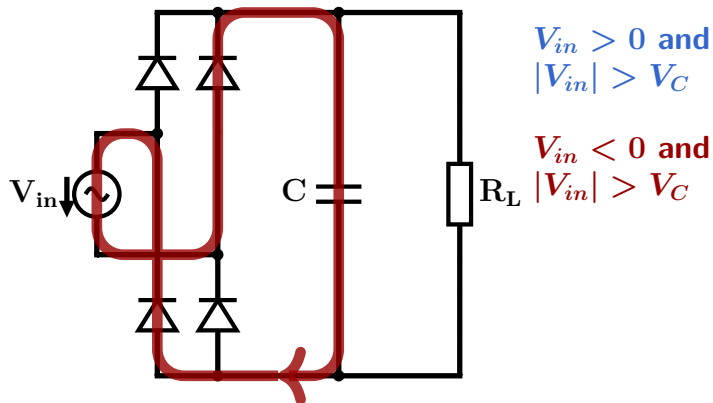
$V_{in} < V_C$:

Capacitor is supplying load

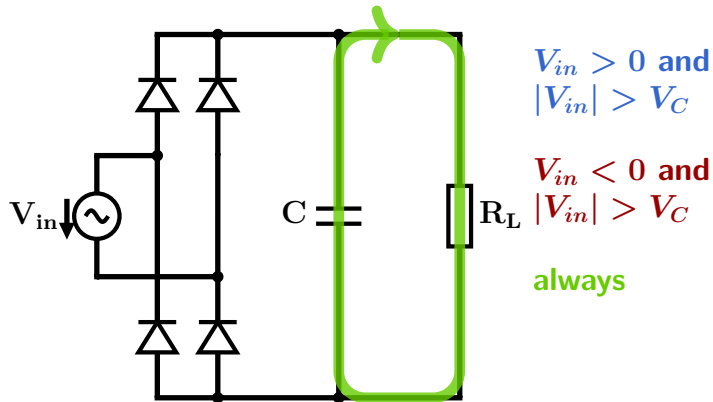
Fullbridge rectifier



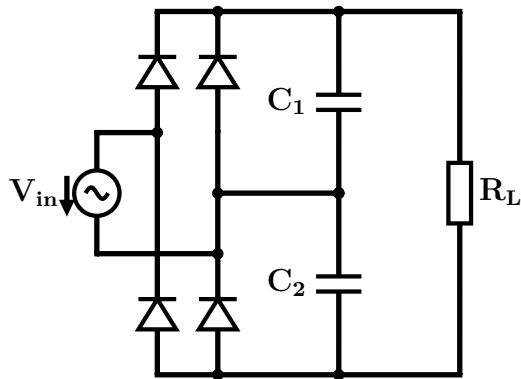
Fullbridge rectifier



Fullbridge rectifier

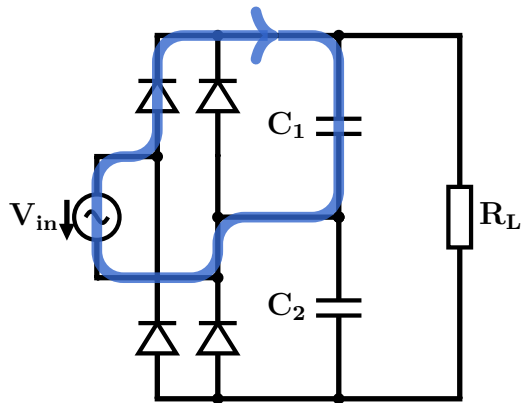


Voltage doubler



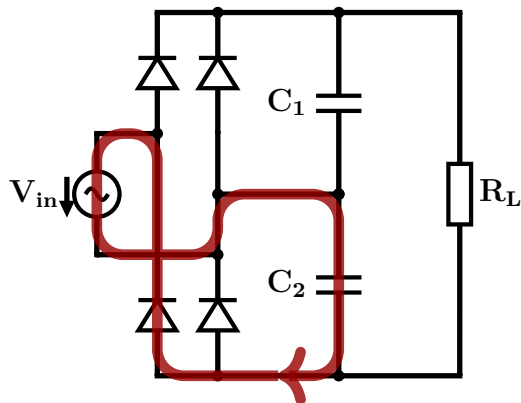
splitting capacitor
into 2 capacitors
+ adding a switch

Voltage doubler



splitting capacitor
into 2 capacitors
+ adding a switch
 $V_{in} > 0$ &
 $|V_{in}| > V_{C1} \Rightarrow$
 V_{C1} charges to $\hat{V}_{in}$

Voltage doubler

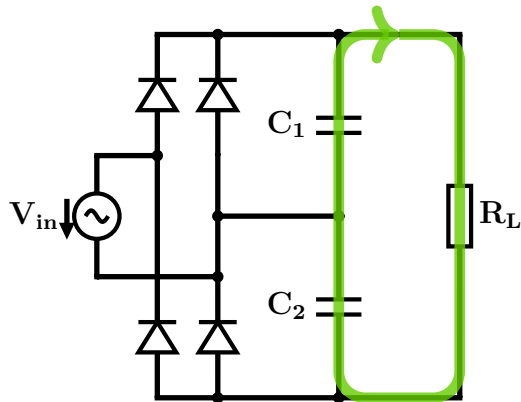


splitting capacitor
into 2 capacitors
+ adding a switch

$V_{in} > 0$ &
 $|V_{in}| > V_{C1} \Rightarrow$
 V_{C1} charges to $\hat{V}_{in}$

$V_{in} < 0$ &
 $|V_{in}| > V_{C2} \Rightarrow$
 V_{C2} charges to $\hat{V}_{in}$

Voltage doubler



splitting capacitor
into 2 capacitors
+ adding a switch

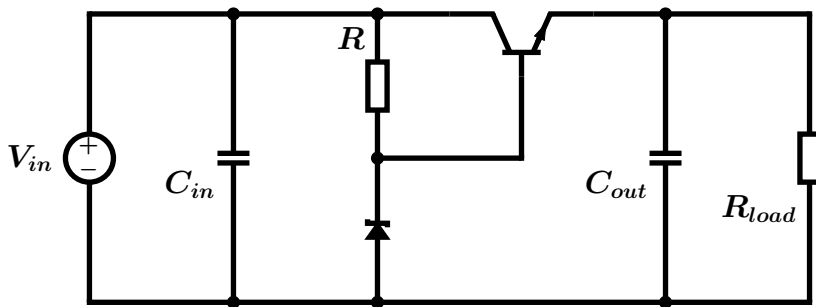
$V_{in} > 0$ &
 $|V_{in}| > V_{C1} \Rightarrow$
 V_{C1} charges to $\hat{V}_{in}$

$V_{in} < 0$ &
 $|V_{in}| > V_{C2} \Rightarrow$
 V_{C2} charges to $\hat{V}_{in}$

always

Linear regulator

e.g. LM7805 for 5 V

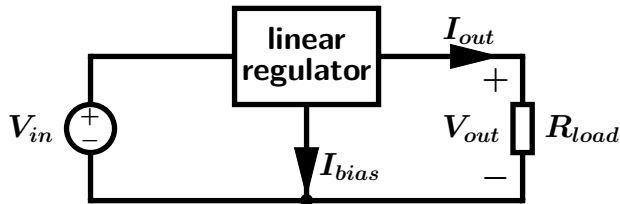


Linear regulator

Exercise 1.2 Low voltage microcontroller supply

A voltage regulator is supplying a microcontroller from a voltage source. The specifications are:

V_{in}	5 V
V_{out}	1.5 V
I_{out}	15 A
R_{load}	0.1 Ω



What is the maximum achievable efficiency of the linear regulator?

Characterization by input and output

DC output

AC output

**DC
input**

switch-mode power supplies

automotive, point-of-load converters (supply microcontrollers, processors, memory, ...), photovoltaic (PV) applications (garden LED lamps)

amplifiers, inverters

audio amplifiers, motordrives, PV-inverters (from solar cells to grid)

**AC
input**

rectifiers

typically used as first stage in grid connected applications, e.g. power factor correction

frequency converters

speed control (low level power tools like drills), phase control (dimming in lighting), rarely used directly - mainly through intermediate DC bus

Section 2

Signals

- **Introduction**
- **Signals**
 - Definitions
 - Superpositions of AC & DC
 - Power Factor
 - Laboratory Exercises for Signals

Conditions

For periodic, i.e. time-repetitive signals only!

IMPORTANT Average Value

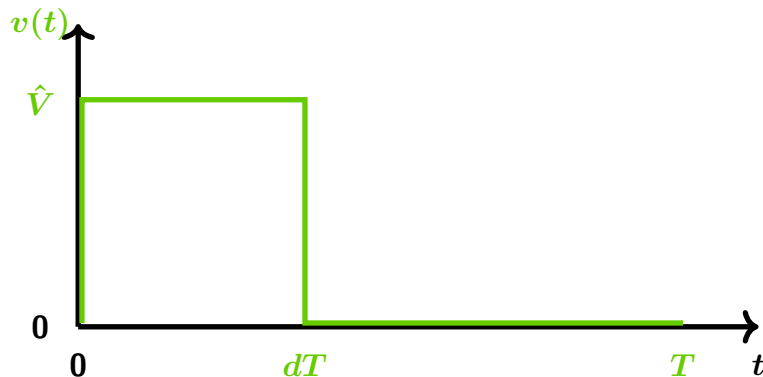
$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt$$

IMPORTANT Root-Mean-Square (RMS) Value

$$V_{RMS} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

IMPORTANT Crest Factor

$$\xi = \frac{\hat{v}}{V_{RMS}}$$

Exercise 2.1 Squarewave with duty cycle d 

What is the mean value V_{avg} , the RMS value V_{RMS} and the crest factor ξ of this repetitive signal, with $v(t) = v(t + T)$?

Superposition of DC & AC

Derivation Elimination of DC by Fourier Series

For periodic signals $v(t) = v(t + T)$,

⇒ Fourier Series can be applied:

$$v(t) = \sum_{n=0}^{\infty} V_n \cos(n\omega t + \varphi_n) \quad (1)$$

⇒ Elimination of DC:

$$v(t) = V_{DC} + \underbrace{\sum_{n=1}^{\infty} V_n \cos(n\omega t + \varphi_n)}_{v_{AC}(t)} \quad (2)$$

$$\Rightarrow v(t) = V_{DC} + v_{AC}(t) \quad (3)$$

Calculating the RMS value of this signal → the periodicity sine waves integrals over a whole period of a sine wave is always zero.



RMS value of signals containing both DC and AC

IMPORTANT

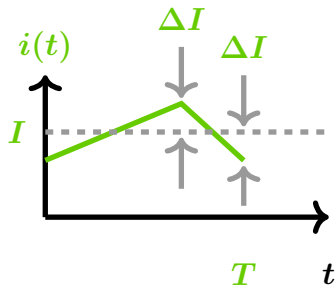
$$V_{RMS} = \sqrt{V_{DC}^2 + V_{AC_{RMS}}^2} \quad (4)$$

Naming

The DC value is also called average, mean, offset or bias point.

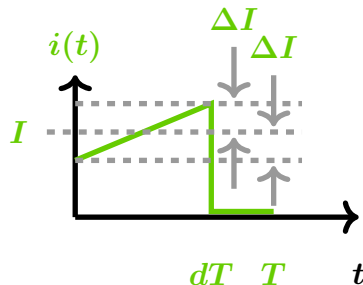
The remaining time dependent part of a signal is often called ripple or AC part.

a) Triangular ripple



$$I_{RMS} = \sqrt{I^2 + \frac{\Delta I^2}{3}}$$

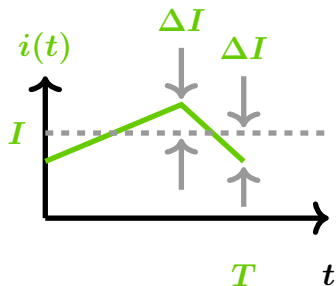
b) Puls with ripple



$$I_{RMS} = \sqrt{d \left(I^2 + \frac{\Delta I^2}{3} \right)}$$

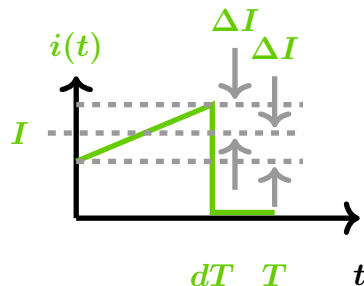
Often occurring waveforms

a) Triangular ripple



$$I_{RMS} = \sqrt{I^2 + \frac{\Delta I^2}{3}}$$

b) Puls with ripple



$$I_{RMS} = \sqrt{d \left(I^2 + \frac{\Delta I^2}{3} \right)}$$

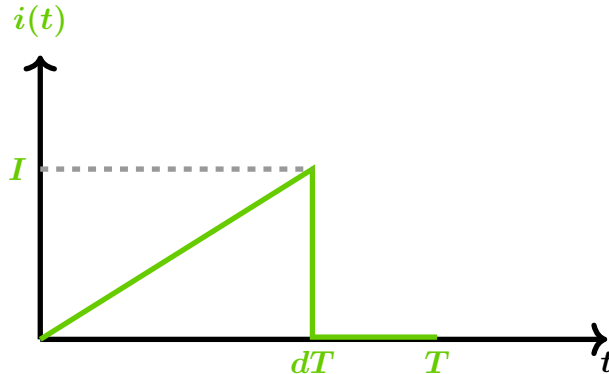
Exercise 2.3 DC Values

What are the DC values the above repetitive signals?

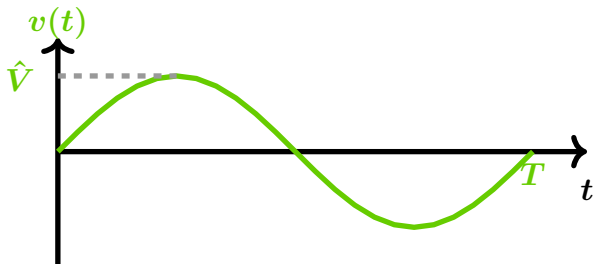
Exercises

What are the mean values and the RMS values of the following repetitive signals?

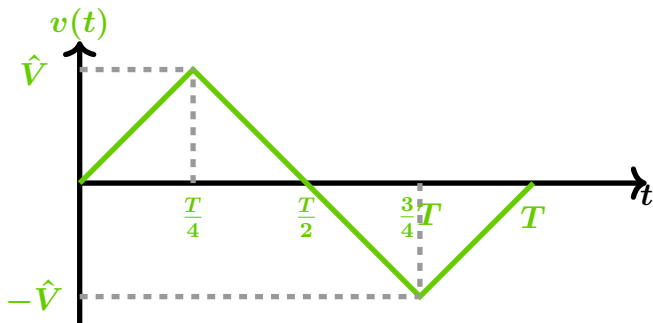
Exercise 2.4 Halfwave sawtooth



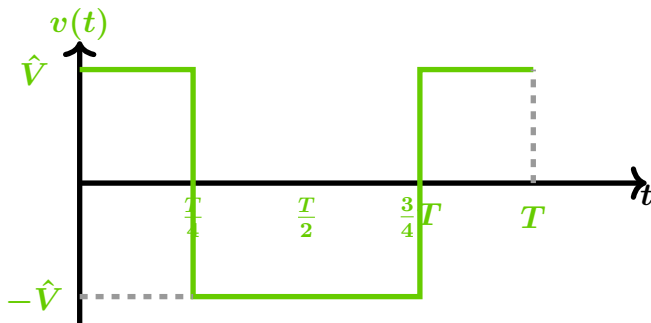
Exercise 2.5 Sine wave $v(t) = \hat{V} \sin(\omega t)$



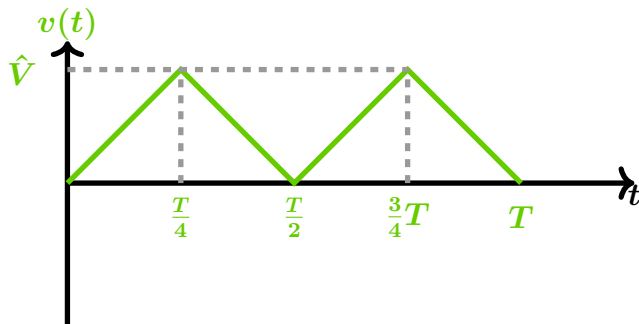
Exercise 2.6 More Triangles, ...



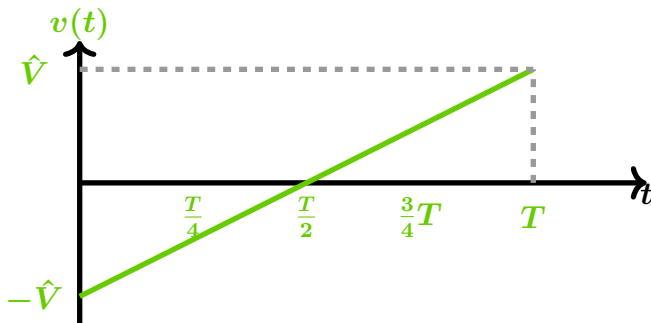
Exercise 2.7 ... another square wave, ...



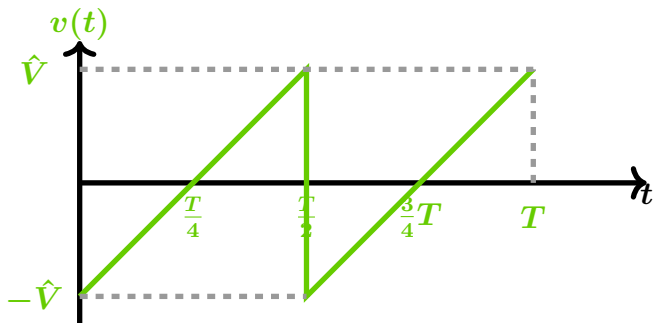
Exercise 2.8 ... two triangles, ...



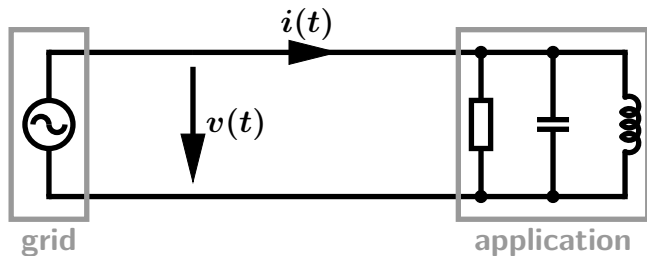
Exercise 2.9 ... a sawtooth, ...



Exercise 2.10 ... and two sawteeth.



AC power source and load



Grid voltage and load current

$$v(t) = \hat{V} \sin(\omega t + \varphi_V) \quad (5)$$

$$i(t) = \hat{I} \sin(\omega t + \varphi_I) \quad (6)$$

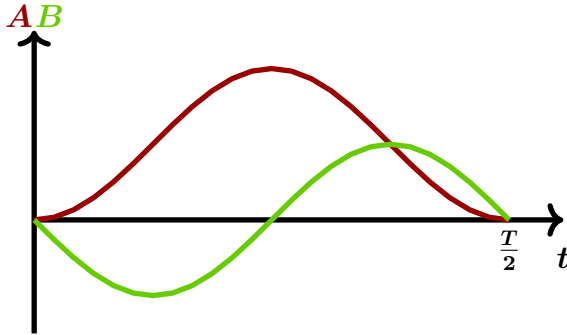
Derivation as product of voltage and current

$$\begin{aligned} p(t) &= v(t)i(t) \\ &= \dots \\ &= \frac{\hat{V}\hat{I}}{2} \left[\left\{ \overbrace{1 - \cos(2\omega t + 2\varphi_V)}^A \right\} \cos(\varphi_V - \varphi_I) \right. \\ &\quad \left. - \underbrace{\sin(2\omega t + 2\varphi_V)}_B \sin(\varphi_V - \varphi_I) \right] \end{aligned} \quad (7)$$

Signals

Partial signals **A** and **B**

look like this:



Deduction

- **A** is always positive

- **B** has no DC part

Power definitions

therefore

$$P = \frac{\hat{V}\hat{I}}{2} (1 - \cos(2\omega t + 2\varphi_V)) \cos(\varphi_V - \varphi_I)$$

$$Q = -\frac{\hat{V}\hat{I}}{2} \sin(2\omega t + 2\varphi_V) \sin(\varphi_V - \varphi_I)$$
(8)

IMPORTANT different parts of power:

- P is called active power or real power,
- Q is called reactive power and
- $\underline{S} = P + jQ$ is called complex power

IMPORTANT power factor

$$\lambda = \frac{P}{S}$$
(9)

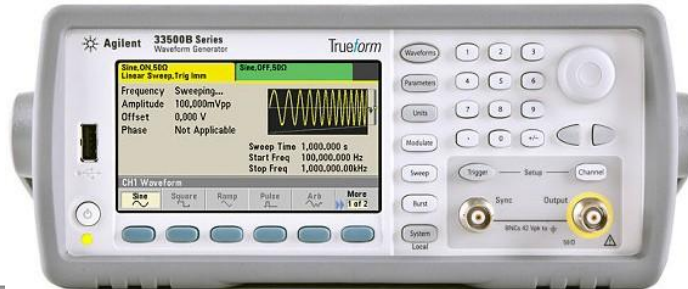


Function generator

Get familiar with the function generator

Turn on and play with it:

- different waveforms,
- amplitude,
- offset,
- frequency.



Signals

Oscilloscope

Get familiar with the oscilloscope

Turn on and play with it:

- channel settings,
- trigger settings,
- cursor,
- measurement settings,
- saving data to disk,
- taking screenshots.



Experimental verifying

of waveforms

Verify the calculations of the waveforms from exercises 2.1, 2.4, 2.5, 2.6, 2.7, 2.8, and 2.9 with the waveform generator and the oscilloscope.

bonus task:

Verify the calculations of the remaining waveforms 2.2, and 2.3, with the arbitrary waveform generator function in the waveform generator and the oscilloscope.

Part II

Components

Section 3

Passive Components

- **Passive Components**

- Resistors
- Inductors
- Capacitors

- **Active Components - Power Semiconductors**

Dependency on material and geometry

$$R = \rho \frac{l}{A} \quad (10)$$

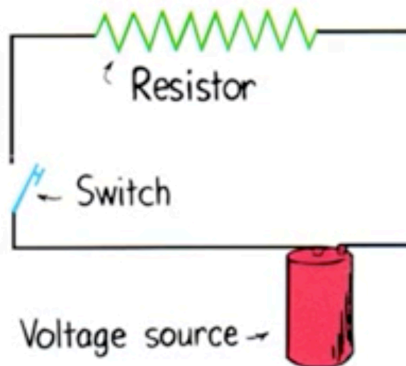
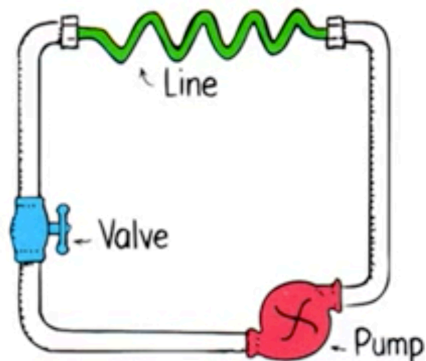
Ohms law

$$V = RI \quad (11)$$

where

- R is the resistance,
- ρ is the resistivity
- l is the length of the conductor
- A is the diameter of the conductor

Analogy



hydraulic $\Leftrightarrow$ electric
water pressure $\Leftrightarrow$ voltage
sand in the water hose $\Leftrightarrow$ resistivity
(flow) current $\Leftrightarrow$ (electrical) current

Energy

$$E = \int p(t)dt = \int v(t)i(t)dt = \frac{1}{R} \int v^2(t)dt \quad (12)$$

Derivation Power

$$P_{avg} = \frac{1}{T}E = \frac{1}{T} \frac{1}{R} \int_0^T v^2(t)dt = \frac{1}{T}R \int_0^T i^2(t)dt \quad (13)$$

Derivation Power for DC

$$P_{avgDC} = \frac{V_{DC}^2}{R} = I_{DC}^2 R \quad (14)$$

Derivation Power for AC

$$P_{avgAC} = \frac{1}{RT} \int_0^T v_{AC}^2(t)dt \quad (15)$$

$$= \frac{R}{T} \int_0^T i_{AC}^2(t)dt \quad (16)$$

Power in a resistor $\Rightarrow$ RMS value of any AC signal

$$V_{DC}^2 = \frac{1}{T} \int_0^T v_{AC}^2(t) dt \qquad I_{DC}^2 = \frac{1}{T} \int_0^T i_{AC}^2(t) dt$$

IMPORTANT Meaning of RMS value of any AC signal

The RMS value of any random signal is equivalent to the DC value of a DC signal, that needs to be applied to a given resistor R to cause the exact same amount of power loss in this resistor.

Analogy



- like the current in a watermill the current in an inductor can not abruptly change
- changing the pressure (voltage) changes the current flow gradually

Faradays induction law

$$N\phi = \int V dt \quad (17)$$

$$\vec{B} = \frac{\phi}{A_{eff}} = \frac{\int V dt}{N A_{eff}} \quad (18)$$

$$\Rightarrow V = N A_{eff} \frac{d\vec{B}}{dt} \quad (19)$$

Amperes law

$$NI = \int_0^{l_{eff}} \vec{H} dl \quad (20)$$

$$\Rightarrow I = \frac{\int \vec{H} dl}{N} \quad (21)$$

Nomenclature

- N number of turns
- ϕ magnetic flux
- $\vec{B}$ flux density
- A_{eff} effective cross section
- l_{eff} effective length
- $\vec{H}$ magnetic field

Ohms law applied to inductors

$$L = \frac{\int V dt}{I} = \frac{\int N A_{eff} \frac{d\vec{B}}{dt} dt}{\frac{\int_0^{l_{eff}} \vec{H} dl}{N}} = N^2 A_{eff} \frac{\vec{B}}{\int_0^{l_{eff}} \vec{H} dl} \quad (22)$$

which gives for homogeneous and linear materials:

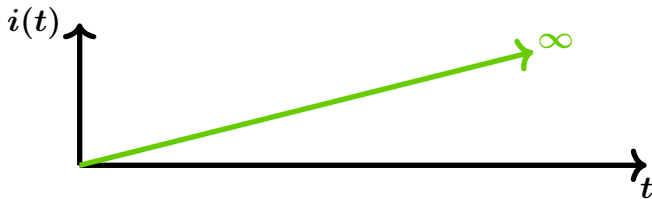
$$L = \underbrace{\frac{\vec{B}}{\vec{H}}}_{\substack{\mu \\ A_L \text{ value}}} \frac{A_{eff}}{l_{eff}} N^2 \quad (23)$$

IMPORTANT Inductance dependency on material and geometry - or the engineering A_L value

$$L = \mu \frac{A_{eff}}{l_{eff}} N^2 = A_L N^2 \quad (24)$$

DC voltage across an inductor

$$i(t) = \frac{\int v(t) dt}{L} = \frac{V_{DC}}{L} \int_0^{\infty} dt = \infty \quad (25)$$



IMPORTANT Never a DC voltage across an inductor!

$$\Rightarrow \int_0^{\infty} v_L(t) \stackrel{!}{=} 0 \quad (26)$$

Derivation Energy in an inductor

$$E_L = \int v_L(t) i_L(t) dt = L \int i_L(t) \frac{di_L}{dt} dt \quad (27)$$

Derivation using the rule for partial integration

$$\int x \frac{dx}{dt} dt = x^2 - \int x \frac{dx}{dt} dt \Rightarrow \int x \frac{dx}{dt} dt = \frac{x^2}{2} \quad (28)$$

IMPORTANT gives the energy stored in an inductor at a given time

$$E_L(t) = \frac{1}{2} L i_L^2(t) \quad (29)$$

Equivalent circuit diagrams of inductors

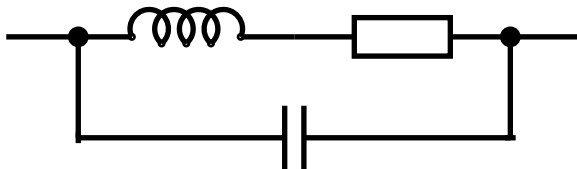
Circuit diagram of an ideal inductor



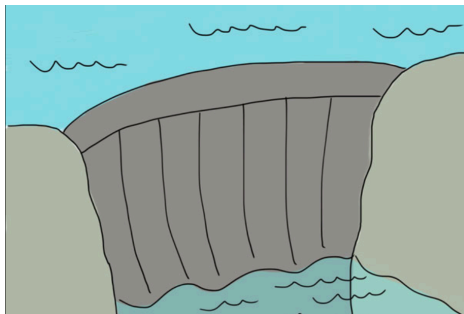
Circuit diagram of an inductor including DC losses



Circuit diagram of an inductor including first resonance



Analogy



- like a dam withstands the pressure from the water, a capacitor sustains voltage
- a current flow changes the water level (voltage)

Gaus law

$$q = \oint_A \vec{D} d\sigma \quad (30)$$

Faradays law

$$V = \oint_d \vec{E} ds \quad (31)$$

Nomenclature

- q charge
- $\vec{D}$ displacement field
- A surface
- σ 2D path along surface
- $\vec{E}$ electric field
- l length of a loop
- s path along loop

Ohms law applied to capacitors

$$C = \frac{\int I dt}{V} = \frac{\int \frac{dq}{dt} dt}{V} = \frac{q}{V} = \frac{\oint_A \vec{D} d\sigma}{\oint_d \vec{E} ds} \quad (32)$$

which gives for homogeneous and linear materials:

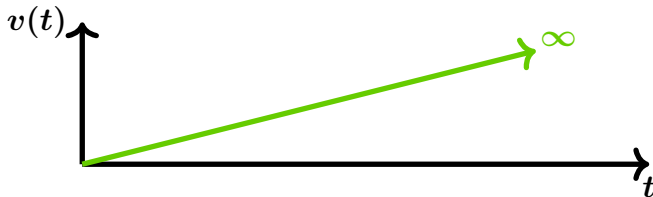
$$C = \frac{\vec{D}}{\underbrace{\vec{E}}_{\epsilon}} \frac{A}{d} \quad (33)$$

IMPORTANT Capacitance dependency on material and geometry

$$C = \frac{q}{V} = \epsilon \frac{A}{d} \quad (34)$$

DC current through a capacitor

$$v(t) = \frac{\int i(t) dt}{C} = \frac{I_{DC}}{C} \int_0^{\infty} dt = \infty \quad (35)$$



IMPORTANT Never a DC current through a capacitor!

$$\Rightarrow \int_0^{\infty} i_C(t) dt \stackrel{!}{=} 0 \quad (36)$$

Derivation Energy in a capacitor

$$E_C = \int v_C(t) i_C(t) dt = C \int v_C(t) \frac{dv_C}{dt} dt \quad (37)$$

Derivation using the rule for partial integration

$$\int x \frac{dx}{dt} dt = x^2 - \int x \frac{dx}{dt} dt \Rightarrow \int x \frac{dx}{dt} dt = \frac{x^2}{2} \quad (38)$$

IMPORTANT gives the energy stored in a capacitor at a given time

$$E_C(t) = \frac{1}{2} C v_C^2(t) \quad (39)$$

Equivalent circuit diagrams of capacitors

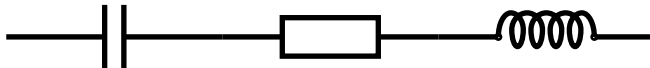
Circuit diagram of an ideal capacitor



Circuit diagram of a capacitor including the equivalent series resistance ESR



Circuit diagram of a capacitor including the ESR & the equivalent series inductance ESL



Exercise 3.1 Impedance of a capacitor

A capacitor has a capacitance $C = 470 \mu\text{F}$ and an equivalent series resistance

of $ESR = 10 \text{ m}\Omega$.



- a What is the magnitude of its impedance at 70 kHz?
- b Use your favourite computer tool to plot the impedance (magnitude & phase) of the capacitor in the frequency range from 1 kHz to 1 MHz! (LTSpice is recommended for this one)
- c Add an equivalent series inductance $ESL = 10 \text{ nH}$ and plot the impedance in comparison to b)!



Exercise 3.1 Impedance of capacitors - bonus tasks

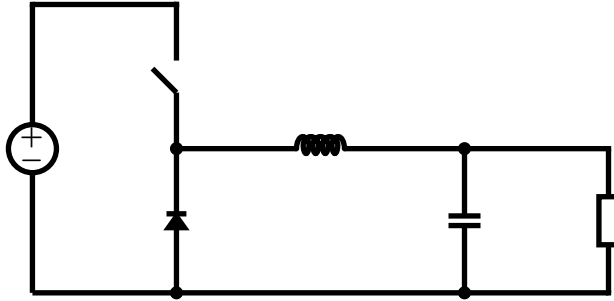
- d** Play with the values and the simulation parameters and see what happens!
(For example $ESR = 1 \text{ m}\Omega$, $ESL = 100 \text{ nH}$; How does the number of datapoints affect the results at the resonance frequency?)
- e** Find some datasheets of different capacitors on the internet, extract the relevant values and enter those in your simulation

Section 4

Active Components - Power Semiconductors

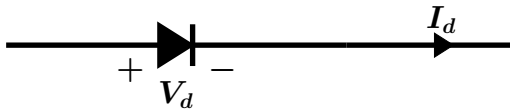
- **Passive Components**
- **Active Components - Power Semiconductors**
 - Diodes
 - MOSFETs
 - Cooling
 - Other Power Semiconductors

Sneak preview of buck converter

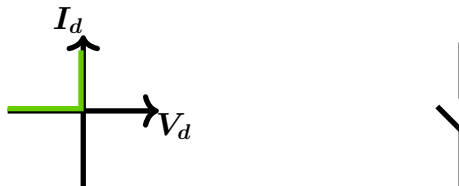


Diodes

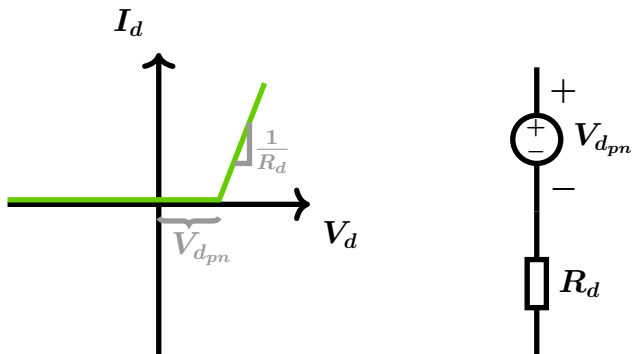
Symbol



Ideal diodes

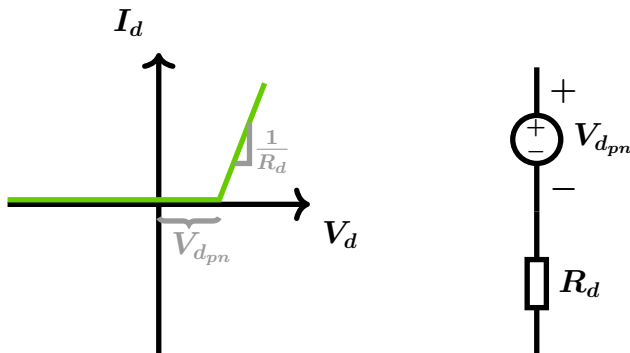


Simple model of diodes



Diodes

Simple model of diodes

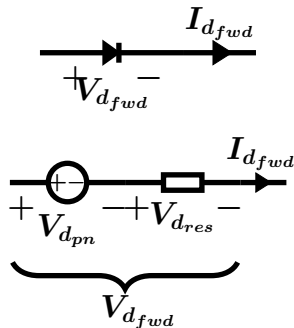


Real power diodes

⇒ datasheet of MUR820Pbf

Losses in power diodes

Conducting diode



Conduction losses

$$E_{d_{cond}} = \int v_d(t) i_d(t) dt \quad (40)$$

which for rectangular signals results in

$$E_{d_{cond}} = V_{d_{fwd}} I_{d_{fwd}} \int_{t_1}^{t_2} dt \quad (41)$$

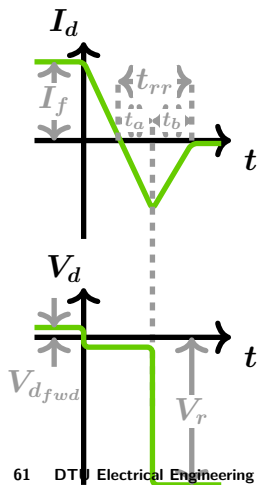
$$= V_{d_{fwd}} I_{d_{fwd}} (t_2 - t_1) \quad (42)$$

and for repetitive signals in

$$P_{d_{cond}} = \frac{E_{d_{cond}}}{T} \quad (43)$$

Losses in power diodes

Reversing diode



Reverse recovery losses

$$E_{dsw} = \int v(t)i(t)dt \quad (44)$$

$$= V_r \int i(t)dt \quad (45)$$

$$= V_r \frac{I_{rrpk} t_b}{2} \quad (46)$$

which gives for repetitive signals:

$$P_{dsw} = \frac{E_{dsw}}{T} = E_{dsw} f_{sw} \quad (47)$$

Total losses in diode

$$P_d = P_{dcond} + P_{dsw} \quad (48)$$

Diode exercise

Exercise 4.1 Static losses

The diode MUR820PbF shall be used in a buck converter that is operating with a duty cycle of $d = 0.25$ and a switching frequency $f_{sw} = 200$ kHz. The converter's average output current is $I_{out} = 8$ A and its input voltage is $V_{in} = 160$ V. At this current level the forward voltage drop across the diode is $V_{f25} = 0.98$ V at 25 °C and $V_{f125} = 0.90$ V at 125 °C.

- a What are the average conduction losses in the diode at 25 °C?
- b What are the average conduction losses in the diode at 125 °C?

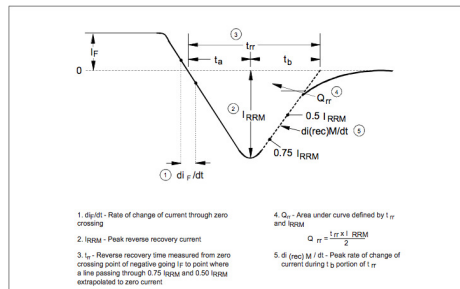
Diode exercise

Exercise 4.1 Dynamic losses at 25 °C

- c** The reverse recovery (as shown in the datasheet figure) current at a junction temperature of 25 °C is peaking at $I_{rr25} = 1.7$ A. It takes $t_{b25} = 10$ ns until the reverse recovery current has returned from its peak to zero. The waveform shape can be assumed to be triangular.

Dynamic Recovery Characteristics @ $T_J = 25^\circ\text{C}$ (unless otherwise specified)

Parameters	Min	Typ	Max	Units	Test Conditions
t_{rr} Reverse Recovery Time	-	-	35	ns	$I_F = 1.0\text{A}$, $dI_F/dt = 50\text{A}/\mu\text{s}$, $V_R = 30\text{V}$
	-	-	25		$I_F = 0.5\text{A}$, $I_R = 1.0\text{A}$, $I_{REC} = 0.25\text{A}$
	-	20	-		$T_J = 25^\circ\text{C}$
	-	34	-		$T_J = 125^\circ\text{C}$
I_{RRM} Peak Recovery Current	-	1.7	-	A	$I_F = 8\text{A}$ $V_R = 160\text{V}$ $dI_F/dt = 200\text{A}/\mu\text{s}$
	-	4.2	-		$T_J = 25^\circ\text{C}$
	-	-	-		$T_J = 125^\circ\text{C}$
Q_{rr} Reverse Recovery Charge	-	23	-	nC	$T_J = 25^\circ\text{C}$
	-	75	-		$T_J = 125^\circ\text{C}$



For the reverse recovery data of diode: t_b can be assumed to be $t_{rr}/2$ How big are the average switching losses at a junction temperature $\vartheta_j = 25^\circ\text{C}$?

Diode exercise

Exercise 4.1 Dynamic losses at 125 °C and total losses

- d** The reverse recovery current at a junction temperature of 125 °C is peaking at $I_{rr125} = 4.2$ A. It takes $t_{b125} = 17$ ns until the reverse recovery current has returned from its peak to zero. The waveform shape can be assumed to be triangular. How big are the average switching losses at a junction temperature $\vartheta_j = 125$ °C?
- e** How big are the total average losses in the diode at a junction temperature $\vartheta_j = 25$ °C?
- f** How big are the total average losses in the diode at a junction temperature $\vartheta_j = 125$ °C?

Measuring a diode

Static characterization of MUR820PbF

- use a DC voltage source and apply a forward voltage (max. 1.0 V) across the diode
- use one multimeter to monitor the current through the diode
- use another multimeter to monitor the case temperature of the diode (stay below 50 °C)
- Plot the V-I graph of the diode & compare to its datasheet
- Plot the V- ϑ graph of the diode in the same diagram (2nd y-axis)



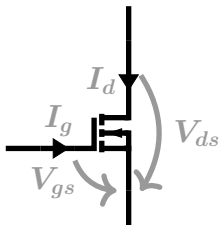
Keep an eye on the
temperature all the time!
Do not touch the diode when hot!



Active Components - Power Semiconductors

MOSFETs

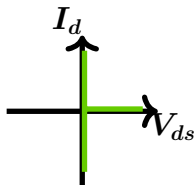
Symbol



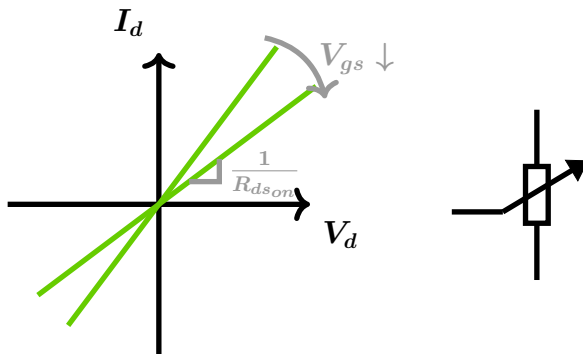
Abbreviation MOSFET

Metal
Oxid
Semiconductor
Field
Effect
Transistor

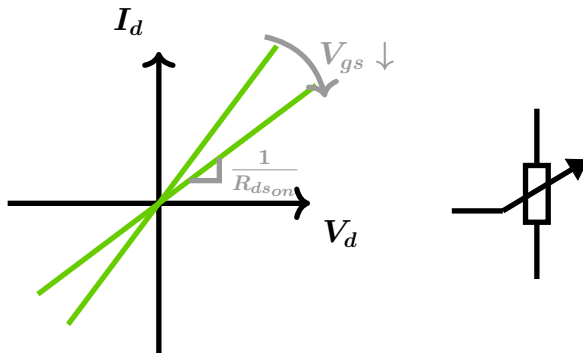
Ideal MOSFETs



Simple model of MOSFETs



Simple model of MOSFETs



Real power MOSFETs

⇒ datasheet of IRF540

Losses in power MOSFETs

Loss mechanisms

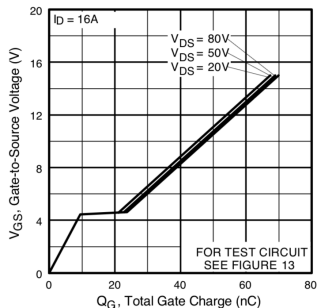
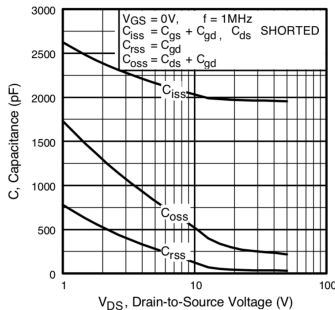
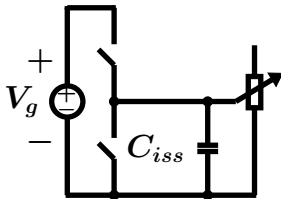
- conduction losses
- switching losses
- gate drive losses

Conduction losses

$$P_{M_{cond}} = R_{ds_{on}} I_{ds_{RMS}}^2 \quad (49)$$

Losses in power MOSFETs

Gate drive circuit

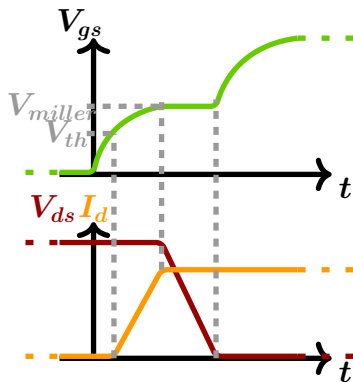


Gate drive losses

$$E_{M_{gate-driv}} = \frac{1}{2} C_g V_g^2 = \frac{1}{2} Q_g V_g \quad (50)$$

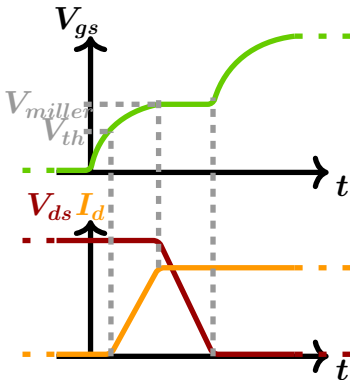
Losses in power MOSFETs

Turn-on switching losses

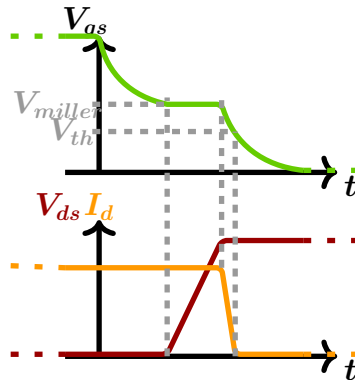


Losses in power MOSFETs

Turn-on switching losses

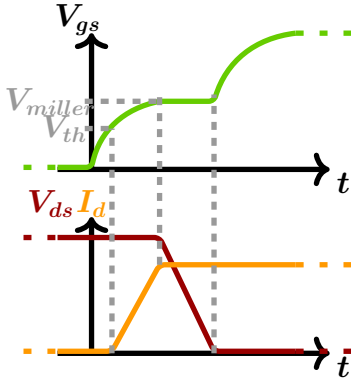


Turn-off switching losses

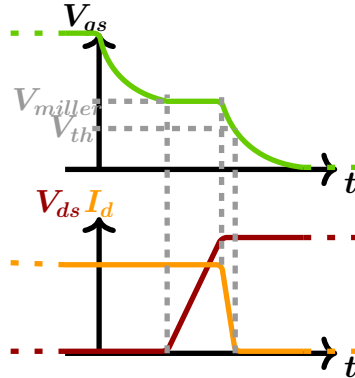


Losses in power MOSFETs

Turn-on switching losses



Turn-off switching losses



Analytical expression for MOSFET switching losses

Modelling MOSFETs



Static characterization of IRF540

- Use a DC voltage source and apply as gate-source voltage V_{gs} within the given limits in the datasheet (hint: steps of 0.2 V are adequate!)
- Use a multimeter to measure the drain-source on-resistance $R_{ds_{on}}$.
- Plot $R_{ds_{on}}$ as a function of V_{gs} & compare to its datasheet.
- What is the threshold voltage V_{th} of the device?
- How are your measurements compared to the ones from other groups?

Mechanical

Si, die, chip
(junction)



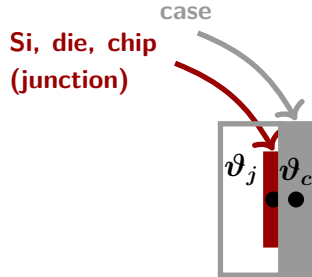
Electrical equivalent

- ϑ_j

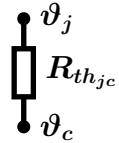
Active Components - Power Semiconductors

Cooling

Mechanical



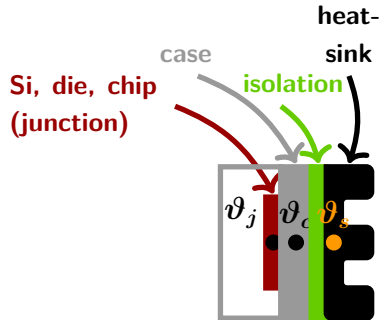
Electrical equivalent



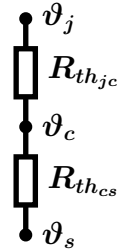
Active Components - Power Semiconductors

Cooling

Mechanical



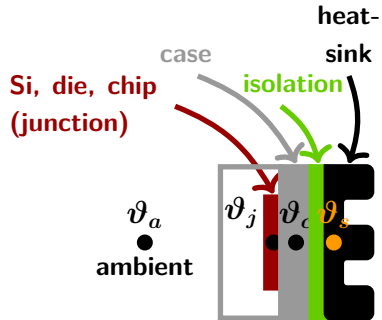
Electrical equivalent



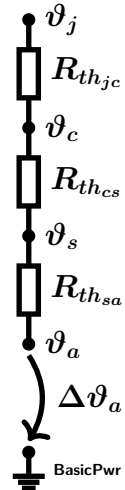
Active Components - Power Semiconductors

Cooling

Mechanical



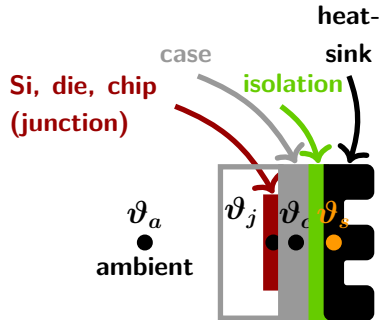
Electrical equivalent



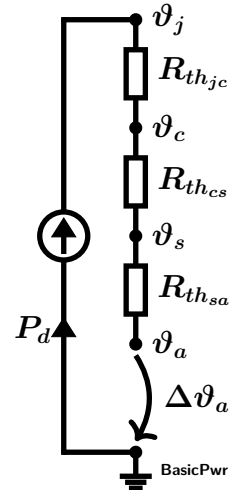
Active Components - Power Semiconductors

Cooling

Mechanical



Electrical equivalent



Power dissipation in die

Physical power P_d modelled as current source!

Cooling

Nomenclature

thermal	→	electric
P_d	→	current
ϑ	→	potential
$\Delta\vartheta$	→	voltage
R_{th}	→	resistance

Other analogies

type	hydraulic	electric	thermal	mechanical
quantity	volume V [m ³]	charge q [C]	heat Q [J]	momentum P [Ns]
potential	pressure p [Pa=J/m ³ =N/m ²]	potential ϕ [V=J/C=W/A]	temperature T [K]	velocity v [m/s]
flux	Volumetric flow rate Φ_V [m ³ /s]	current I [A=C/s]	heat transfer rate $\dot{Q}$ [J/s]	force F [N]
flux density	velocity v [m/s]	current density j [C/(m ² ·s) = A/m ²]	heat flux $\dot{Q}''$ [W/m ²]	stress σ [N/m ² = Pa]
linear model	Poiseuille's law $\Phi_V = \frac{\pi r^4}{8\eta} \frac{\Delta p^*}{\ell}$	Ohm's law $j = -\sigma \nabla \phi$	Fourier's law $\dot{Q}'' = \kappa \nabla T$	dashpot $\sigma = c \Delta v$

Cooling

Exercise 4.2 Cooling of MUR820PbF

The diode MUR820 dissipates 6.2 W. The thermal resistance stated in the datasheet are $R_{th_{ja}} = 50 \frac{\text{K}}{\text{W}}$ and $R_{th_{jc}} = 3 \frac{\text{K}}{\text{W}}$.

- a How high is the junction temperature (above ambient) if no cooling is applied?
- b How high is the junction temperature (above ambient) if perfect cooling $R_{th_{ca}} = 0 \frac{\text{K}}{\text{W}}$ is applied?

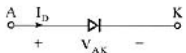
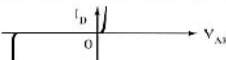
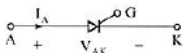
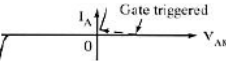
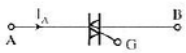
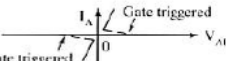
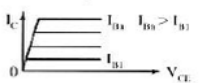
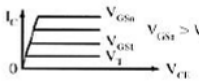
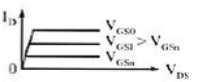
Cooling

Exercise 4.3 Ambient temperature specification

A power semiconductor has a total power dissipation of $P_{d_{total}} = 6.5 \text{ W}$. Its maximum operating junction temperature is $\vartheta_{j_{max}} = 175 \text{ }^{\circ}\text{C}$. The thermal resistances of its package are given as $R_{th_{ja}} = 50 \frac{\text{K}}{\text{W}}$ and $R_{th_{jc}} = 3.0 \frac{\text{K}}{\text{W}}$.

- a What is the maximum surrounding (ambient) temperature, the semiconductor can operate in, with no further cooling attached to its case?
- b What is the maximum surrounding (ambient) temperature, the semiconductor can operate in, with perfect cooling attached to its case?

Other power semiconductors

Devices	Symbols	Characteristics
Diode		
Thyristor		
TRIAC		
NPN BJT		
IGBT		
N-Channel MOSFET		

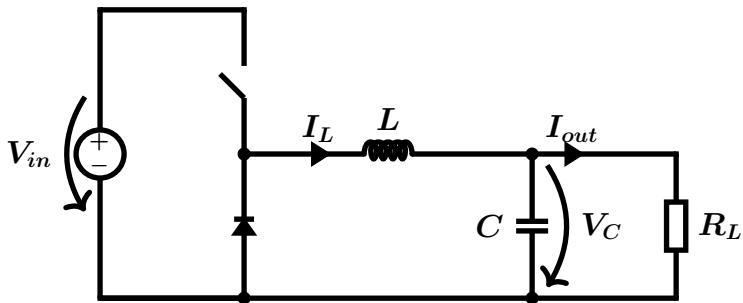
Part III

Switch-Mode Power Supplies

Section 5

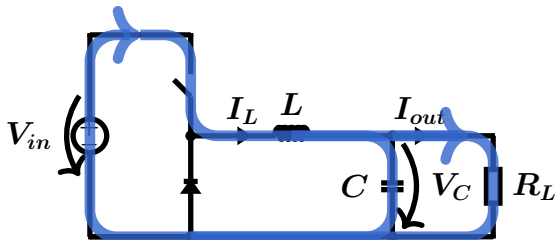
Nonisolated Converters

- **Nonisolated Converters**
 - Buck Converter
 - Boost Converter
 - Buck-boost Converter
- **Parameters of Converters**

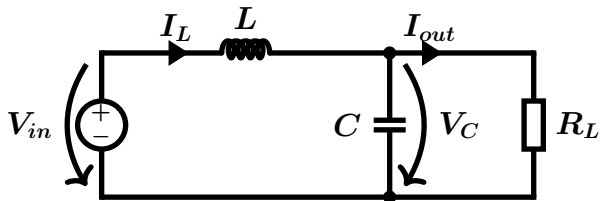
Circuit diagram

Buck - operation modes

$$0 < t < dT$$

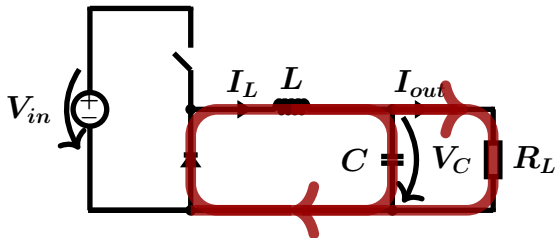


Equivalent circuit diagram during $0 < t < dT$

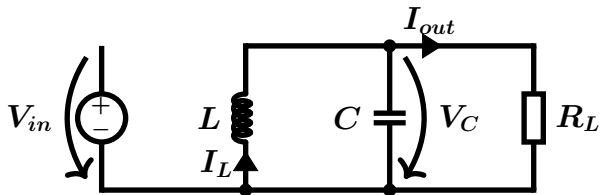


Buck - operation modes

$$dT < t < T$$



Equivalent circuit diagram during $dT < t < T$



Buck - averaging

Assumptions

- steady-state operation
- signals are ripple free
- repetitive operation
- linearization around bias point
- all signals are either DC or rectangular
- all time-derivatives are zero

⇒ analyzing the average values

Nonisolated Converters

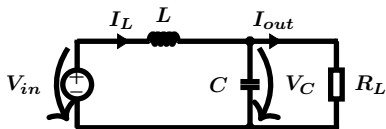
Buck - averaging

Derivation $0 < t < dT$

inductor voltage:

capacitor current:

Equivalent circuit diagram during
 $0 < t < dT$

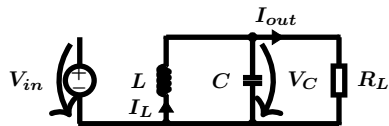


Derivation $dT < t < T$

inductor voltage:

capacitor current:

Equivalent circuit diagram during
 $dT < t < T$



Nonisolated Converters

Buck - averaging

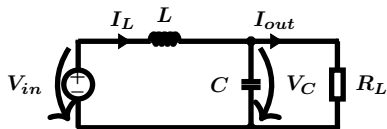
Derivation $0 < t < dT$

inductor voltage:

$$V_{in} - V_C$$

capacitor current:

Equivalent circuit diagram during
 $0 < t < dT$

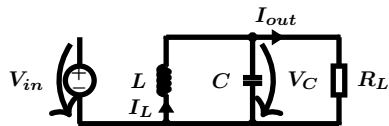


Derivation $dT < t < T$

inductor voltage:

capacitor current:

Equivalent circuit diagram during
 $dT < t < T$



Nonisolated Converters

Buck - averaging

Derivation $0 < t < dT$

inductor voltage:

$$V_{in} - V_C$$

capacitor current:

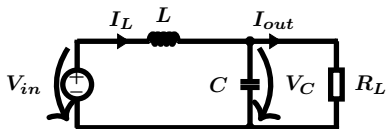
Derivation $dT < t < T$

inductor voltage:

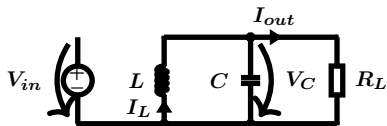
$$-V_C$$

capacitor current:

Equivalent circuit diagram during
 $0 < t < dT$



Equivalent circuit diagram during
 $dT < t < T$



Nonisolated Converters

Buck - averaging

Derivation $0 < t < dT$

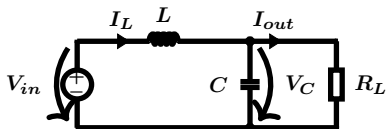
inductor voltage:

$$V_{in} - V_C$$

capacitor current:

$$I_L - I_{out}$$

Equivalent circuit diagram during
 $0 < t < dT$



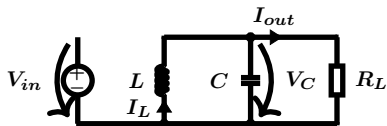
Derivation $dT < t < T$

inductor voltage:

$$-V_C$$

capacitor current:

Equivalent circuit diagram during
 $dT < t < T$



Nonisolated Converters

Buck - averaging

Derivation $0 < t < dT$

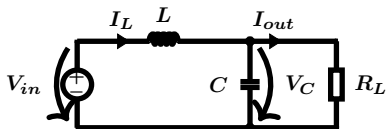
inductor voltage:

$$V_{in} - V_C$$

capacitor current:

$$I_L - I_{out}$$

Equivalent circuit diagram during
 $0 < t < dT$



Derivation $dT < t < T$

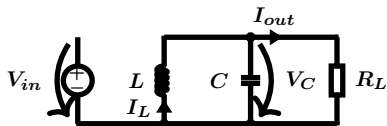
inductor voltage:

$$-V_C$$

capacitor current:

$$I_L - I_{out}$$

Equivalent circuit diagram during
 $dT < t < T$



Buck - averaging

Derivation $0 < t < dT$

inductor voltage:

$$V_{in} - V_C$$

capacitor current:

$$I_L - I_{out}$$

Derivation $dT < t < T$

inductor voltage:

$$-V_C$$

capacitor current:

$$I_L - I_{out}$$

Derivation average inductor voltage & capacitor current

$$V_{L_{avg}} = dT(V_{in} - V_C) - (1 - d)TV_C \stackrel{!}{=} 0$$

$$I_{C_{avg}} = dT(I_L - I_{out}) + (1 - d)T(I_L - I_{out}) \stackrel{!}{=} 0$$

Nonisolated Converters

Buck - averaging

Derivation $0 < t < dT$

inductor voltage:

$$V_{in} - V_C$$

capacitor current:

$$I_L - I_{out}$$

Derivation $dT < t < T$

inductor voltage:

$$-V_C$$

capacitor current:

$$I_L - I_{out}$$

Derivation average inductor voltage & capacitor current

$$V_{L_{avg}} = dT(V_{in} - V_C) - (1 - d)TV_C \stackrel{!}{=} 0$$

$$I_{C_{avg}} = dT(I_L - I_{out}) + (1 - d)T(I_L - I_{out}) \stackrel{!}{=} 0$$

IMPORTANT DC-transfer function of buck

$$\frac{V_C}{V_{in}} = d$$

$$I_{out} = I_L$$

(51)

Black box approach

assumption: efficiency $\eta = 100\%$

$$\frac{I_{in}}{I_{out}} = \frac{\frac{P}{V_{in}}}{\frac{P}{V_{out}}} = \frac{V_{out}}{V_{in}} = d$$

IMPORTANT Input current

$$I_{in} = dI_{out} = dI_L \quad (52)$$

Superposition of ripple = adding AC

Derivation $0 < t < dT$

inductor current ripple:

$$i_L(t) = \frac{\int_0^{dT} (V_{in} - V_C) \delta t}{L} + i_L(0) = dT \frac{V_{in} - V_C}{L} + i_L(0)$$



Superposition of ripple = adding AC

Derivation $0 < t < dT$

inductor current ripple:

$$i_L(t) = \frac{\int_0^{dT} (V_{in} - V_C) \delta t}{L} + i_L(0) = dT \frac{V_{in} - V_C}{L} + i_L(0)$$

capacitor voltage ripple:

$$\begin{aligned} v_C(t) &= \frac{\int_0^{dT} (i_L(t) - I_{out}) \delta t}{C} + v_C(0) \\ &= \frac{\int_0^{dT} (\bar{i}_L(t) + \tilde{i}_L(t) - I_{out}) \delta t}{C} + v_C(0) \end{aligned}$$

$$= \underbrace{\frac{\int_0^{dT} \tilde{i}_L(t) \delta t}{C}}_{\tilde{v}_C(t)} + v_C(0)$$

$$\begin{aligned} \tilde{v}_C(t) &= \frac{\int_0^{dT} \int_0^{dT} (V_{in} - V_C) \delta t \delta t}{LC} + \frac{\int_0^{dT} i_L(0) \delta t}{C} \\ &= \frac{V_{in} - V_C}{2LC} t^2 + \frac{i_L(0)}{C} t \end{aligned}$$



Superposition of ripple = adding AC

Derivation $dT < t < T$

inductor current ripple:

$$i_L(t) = \frac{-\int_{dT}^T V_C \delta t}{L} + i_L(dT) = -(1-d)T \frac{V_C}{L} + i_L(dT)$$

Superposition of ripple = adding AC

Derivation $dT < t < T$

inductor current ripple:

$$i_L(t) = \frac{-\int_{dT}^T V_C \delta t}{L} + i_L(dT) = -(1-d)T \frac{V_C}{L} + i_L(dT)$$

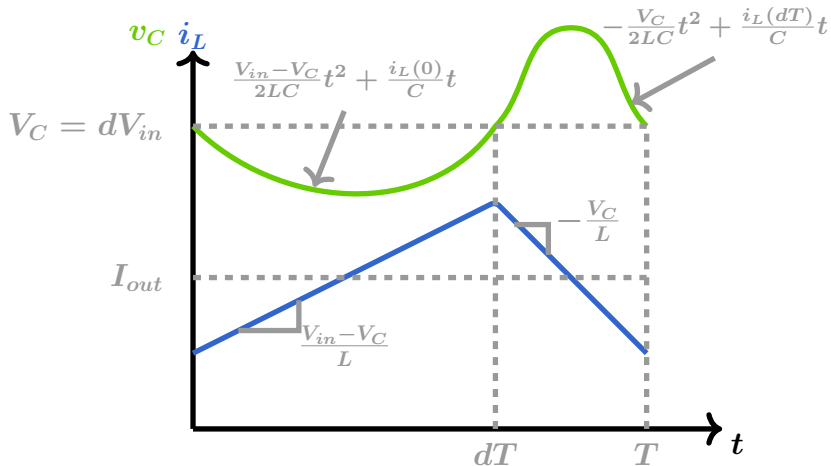
capacitor voltage ripple:

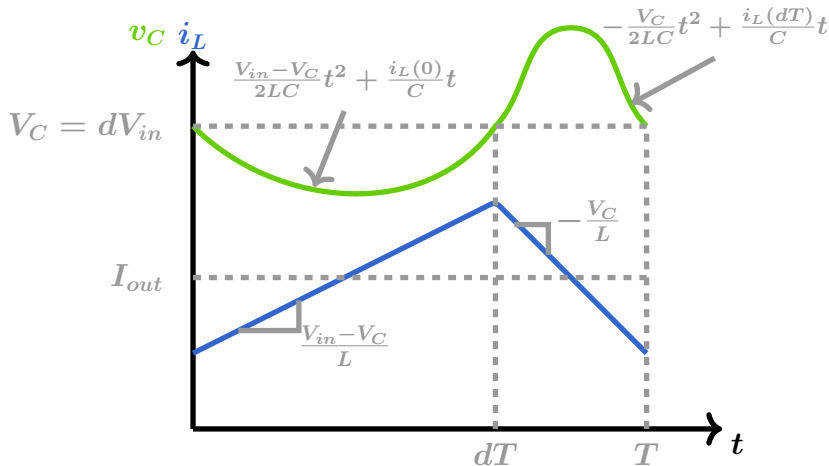
...

$$\tilde{v}_C(t) = -\frac{V_C}{2LC}t^2 + \frac{i_L(dT)}{C}t$$

Buck

Signals

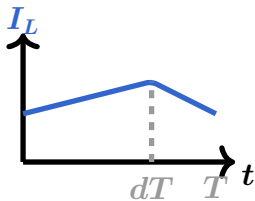




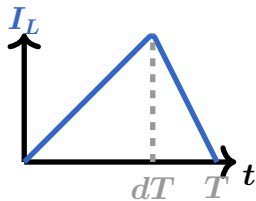
When $I_L > 0 \Rightarrow$ Continuous Conduction Mode (CCM)

Buck - all operating modes

Continuous Conduction Mode (CCM)



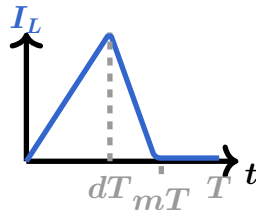
Boundary Conduction Mode (BCM)



also

called Critical Conduction Mode (CrCM) (sometimes abbreviated CCM)

Discontinuous Conduction Mode (DCM)



analysis above not valid here! (later in course!)

Buck - Exercise

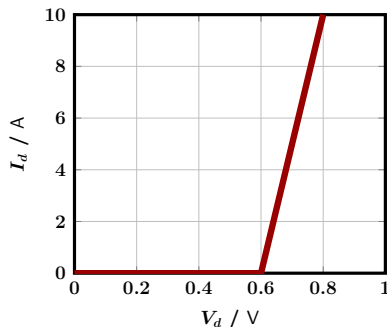
Exercise 5.1 Ideal Buck...

An ideal buck converter is operating from $V_{in} = 30 \text{ V}$ and delivers $I_{out} = 10 \text{ A}$. The inductor current is constant.

- a What is the output voltage if the duty cycle $d = 0.175$?
- b Sketch the current in the active power semiconductors.

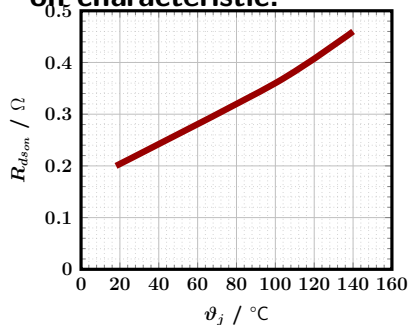
Exercise 5.1 + diode losses

- c How big are the conduction losses in the diode for this VI-characteristic:



Exercise 5.1 + FET losses

- d How big are the conduction losses in the MOSFET for junction temperature $\vartheta_j = 100^\circ\text{C}$ for this on-characteristic:



Buck - Exercise

From now on assume, that the total losses in the MOSFET are

$$P_{d_{MOSFET}} = 9.3 \text{ W.}$$

Exercise 5.1 dimension heatsink for MOSFET

- e The junction temperature of the MOSFET shall be kept at $\vartheta_j = 100 \text{ }^\circ\text{C}$. The inner thermal resistance of the package is $R_{th_{jc}} = 3 \frac{\text{K}}{\text{W}}$ and the maximum ambient temperature is $\vartheta_a = 40 \text{ }^\circ\text{C}$. What is the maximum thermal resistance $R_{th_{ca}}$ of the heatsink?
- f What is the maximum thermal resistance of another heatsink, if both the MOSFET and the diode are mounted to the same heatsink (same conditions as above, diode switching losses are ignored)?

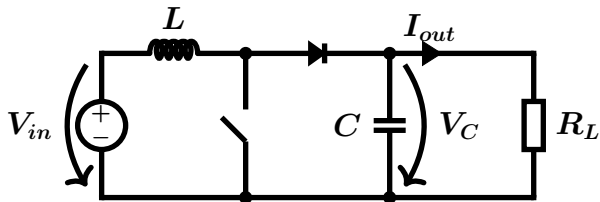
Buck - Exercise

Exercise 5.2 Another ideal buck

An ideal buck converter is switching with $f_{sw} = 100$ kHz. The input voltage $V_{in} = 5$ V and the load is modelled with a resistance $R_L = 0.1 \Omega$. Inductor current and capacitor voltage do not have ripple.

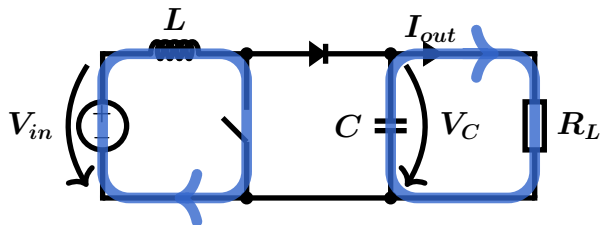
- a What is the maximum theoretical efficiency η ? Why?
- b What is the required duty cycle d to get an output voltage $V_{out} = 1.5$ V?
- c After the input voltage rose, d gets adjusted to 25 % to keep the same output voltage. What is the peak input current $\hat{I}_{in}$?
- d What is the average value of the input current $\bar{I}_{in}$?
- e What is the RMS current of the input current $I_{in_{RMS}}$?
- f Draw the diode current.
- g Everything else unchanged, the MOSFET is now modelled with an on-resistance of $R_{DS_{on}} = 10$ m Ω . How much power is dissipated in the MOSFET?



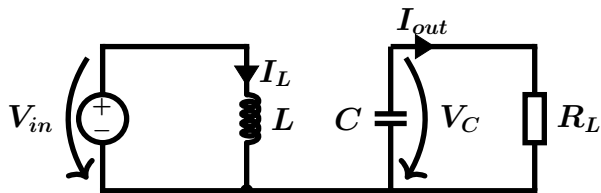
Circuit diagram

Boost - operation modes

$$0 < t < dT$$

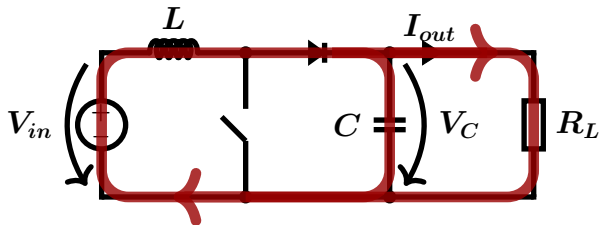


Equivalent circuit diagram during $0 < t < dT$

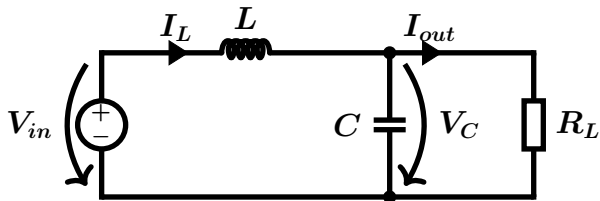


Boost - operation modes

$$dT < t < T$$



Equivalent circuit diagram during $dT < t < T$



Nonisolated Converters

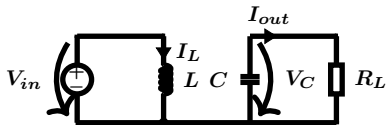
Boost - averaging

Derivation $0 < t < dT$

inductor voltage:

capacitor current:

Equivalent circuit diagram during $0 < t < dT$

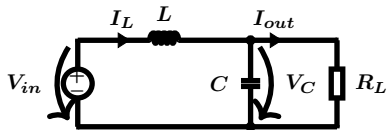


Derivation $dT < t < T$

inductor voltage:

capacitor current:

Equivalent circuit diagram during $dT < t < T$



Boost - averaging

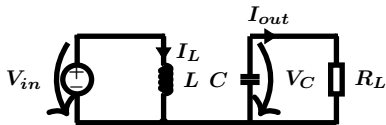
Derivation $0 < t < dT$

inductor voltage:

$$V_{in}$$

capacitor current:

Equivalent circuit diagram during $0 < t < dT$

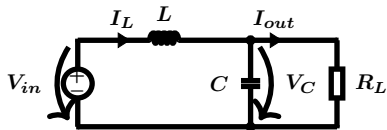


Derivation $dT < t < T$

inductor voltage:

capacitor current:

Equivalent circuit diagram during $dT < t < T$



Nonisolated Converters

Boost - averaging

Derivation $0 < t < dT$

inductor voltage:

$$V_{in}$$

capacitor current:

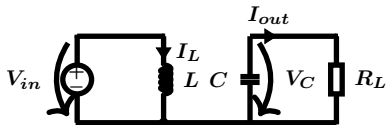
Derivation $dT < t < T$

inductor voltage:

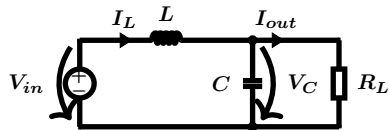
$$V_{in} - V_C$$

capacitor current:

Equivalent circuit diagram during
 $0 < t < dT$



Equivalent circuit diagram during
 $dT < t < T$



Boost - averaging

Derivation $0 < t < dT$

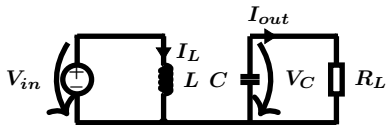
inductor voltage:

$$V_{in}$$

capacitor current:

$$-I_{out}$$

Equivalent circuit diagram during $0 < t < dT$



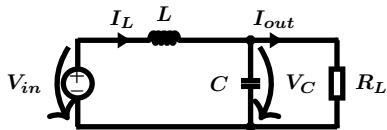
Derivation $dT < t < T$

inductor voltage:

$$V_{in} - V_C$$

capacitor current:

Equivalent circuit diagram during $dT < t < T$



Boost - averaging

Derivation $0 < t < dT$

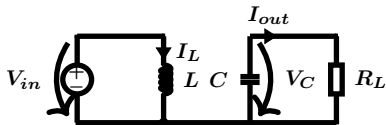
inductor voltage:

$$V_{in}$$

capacitor current:

$$-I_{out}$$

Equivalent circuit diagram during
 $0 < t < dT$



Derivation $dT < t < T$

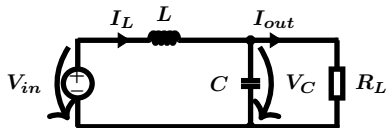
inductor voltage:

$$V_{in} - V_C$$

capacitor current:

$$I_L - I_{out}$$

Equivalent circuit diagram during
 $dT < t < T$



Boost - averaging

Derivation $0 < t < dT$

inductor voltage:

$$V_{in}$$

capacitor current:

$$-I_{out}$$

Derivation $dT < t < T$

inductor voltage:

$$V_{in} - V_C$$

capacitor current:

$$I_L - I_{out}$$

Derivation average inductor voltage & capacitor current

$$V_{L_{avg}} = dTV_{in} + (1 - d)T(V_{in} - V_C) \stackrel{!}{=} 0$$

$$V_{in} = (1 - d)V_C$$

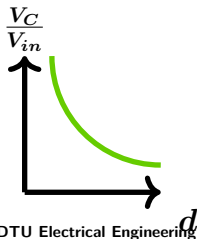
$$I_{C_{avg}} = -dT I_{out} + (1 - d)T(I_L - I_{out}) \stackrel{!}{=} 0$$

$$-I_{out} = -(1 - d)I_L$$

IMPORTANT DC-transfer function of boost

$$\frac{V_C}{V_{in}} = \frac{1}{1-d} \quad (53)$$

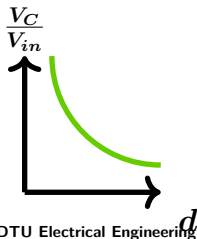
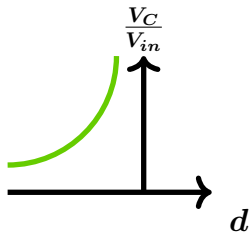
$$\frac{I_L}{I_{out}} = \frac{1}{1-d} \quad (54)$$

 $1/d$ 

IMPORTANT DC-transfer function of boost

$$\frac{V_C}{V_{in}} = \frac{1}{1-d} \quad (53)$$

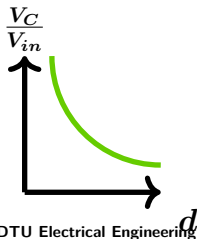
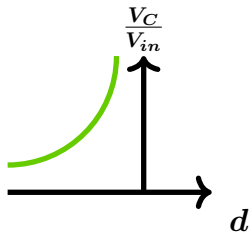
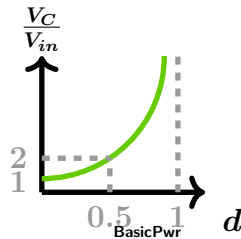
$$\frac{I_L}{I_{out}} = \frac{1}{1-d} \quad (54)$$

 $1/d$  $1/(1-d)$ 

IMPORTANT DC-transfer function of boost

$$\frac{V_C}{V_{in}} = \frac{1}{1 - d} \quad (53)$$

$$\frac{I_L}{I_{out}} = \frac{1}{1 - d} \quad (54)$$

 $1/d$  **$1/-d$**  **$1/(1 - d)$** 

Boost

Black box approach

assumption: efficiency $\eta = 100\%$

$$I_{in} = \frac{V_{out}}{V_{in}} I_{out} = \frac{1}{1-d} I_{out}$$

with

$$I_L = \frac{1}{1-d} I_{out} \Rightarrow I_{out} = (1-d) I_L$$

IMPORTANT Input current

$$I_{in} = I_L \quad (55)$$

Superposition of ripple = adding AC

Derivation $0 < t < dT$

inductor current ripple:

$$i_L(t) = \frac{\int_0^{dT} (V_{in}) \delta t}{L} + i_L(0) = dT \frac{V_{in}}{L} + i_L(0)$$



Superposition of ripple = adding AC

Derivation $0 < t < dT$

inductor current ripple:

$$i_L(t) = \frac{\int_0^{dT} (V_{in}) \delta t}{L} + i_L(0) = dT \frac{V_{in}}{L} + i_L(0)$$

capacitor voltage ripple:

$$v_C(t) = \frac{-\int_0^{dT} I_{out} \delta t}{C} + v_C(0) = -dT \frac{I_{out}}{C} + v_C(0)$$



Superposition of ripple = adding AC

Derivation $dT < t < T$

inductor current ripple:

$$i_L(t) = \frac{\int_{dT}^T (V_{in} - V_C) \delta t}{L} + i_L(dT) = dT \frac{V_{in} - V_C}{L} + i_L(dT)$$



Superposition of ripple = adding AC

Derivation $dT < t < T$

inductor current ripple:

$$i_L(t) = \frac{\int_{dT}^T (V_{in} - V_C) \delta t}{L} + i_L(dT) = dT \frac{V_{in} - V_C}{L} + i_L(dT)$$

capacitor voltage ripple:

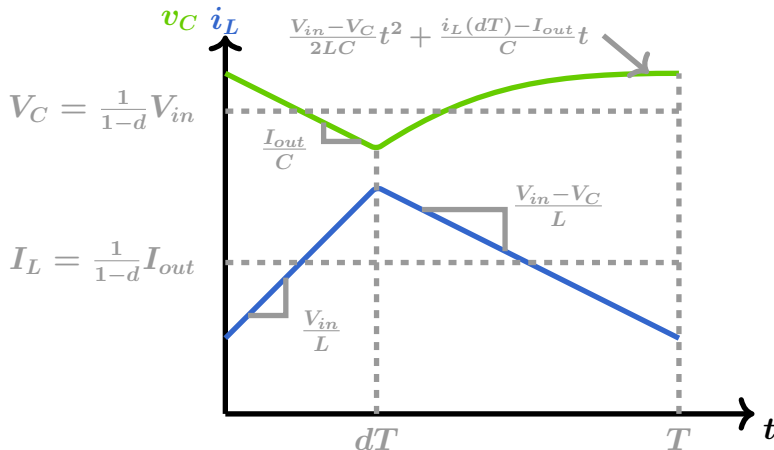
$$\begin{aligned} v_C(t) &= \frac{\int_{dT}^T (i_L(t) - I_{out}) \delta t}{C} + v_C(dT) \\ &= \underbrace{\frac{\int_{dT}^T (i_L(t) - I_{out}) \delta t}{C}}_{\tilde{v}_C(t)} + v_C(dT) \end{aligned}$$

$$\begin{aligned} \tilde{v}_C(t) &= \frac{\int_{dT}^T \int_{dT}^T (V_{in} - V_C) \delta t \delta t}{LC} + \frac{\int_{dT}^T (i_L(dT) - I_{out}) \delta t}{C} \\ &= \frac{V_{in} - V_C}{2LC} t^2 + i_L(dT) - \frac{I_{out}}{C} t \end{aligned}$$



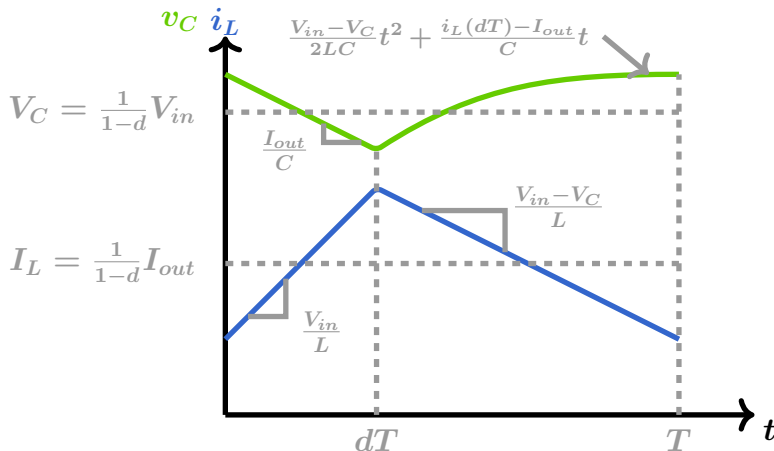
Boost

Signals



Boost

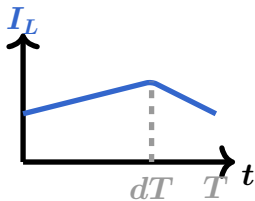
Signals



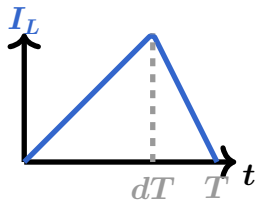
When $I_L > 0 \Rightarrow$ Continuous Conduction Mode (CCM)

Boost - all operating modes

Continuous Conduction Mode (CCM)



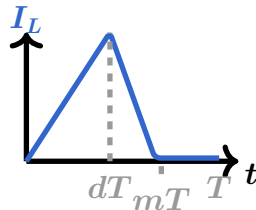
Boundary Conduction Mode (BCM)



also

called Critical
Conduction Mode
(CrCM) (sometimes
abbreviated CCM)

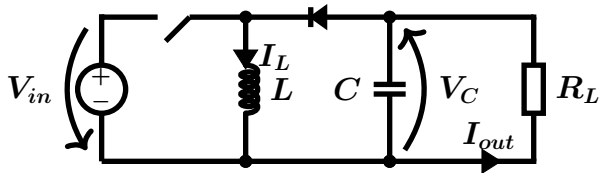
Discontinuous Conduction Mode (DCM)



analysis above not valid
here! (later in course!)

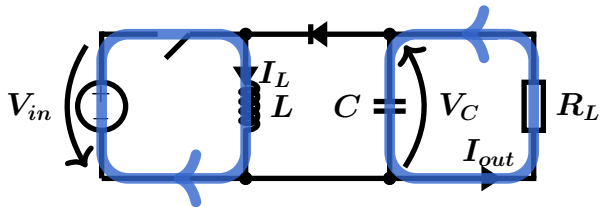
Buck-boost

Circuit diagram

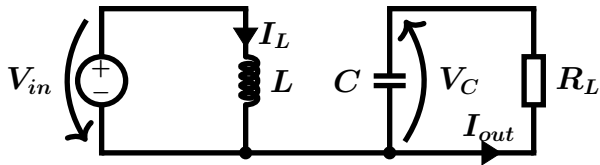


Buck-boost - operation modes

$$0 < t < dT$$

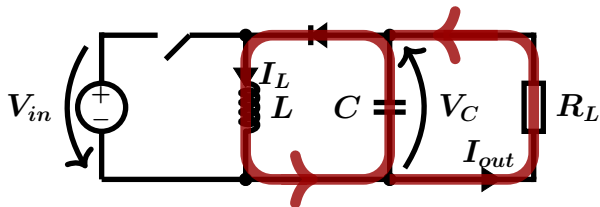


Equivalent circuit diagram during $0 < t < dT$

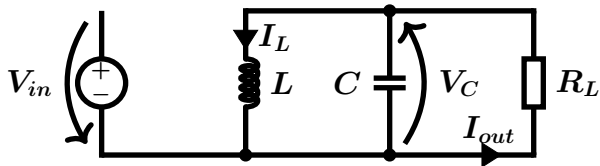


Buck-boost - operation modes

$$dT < t < T$$



Equivalent circuit diagram during $dT < t < T$



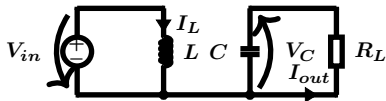
Buck-boost - averaging

Derivation $0 < t < dT$

inductor voltage:

capacitor current:

Equivalent circuit diagram during $0 < t < dT$

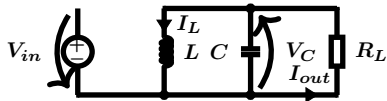


Derivation $dT < t < T$

inductor voltage:

capacitor current:

Equivalent circuit diagram during $dT < t < T$



Buck-boost - averaging

Derivation $0 < t < dT$

inductor voltage:

$$V_{in}$$

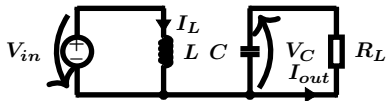
capacitor current:

Derivation $dT < t < T$

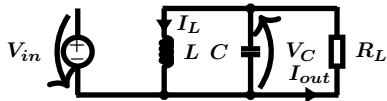
inductor voltage:

capacitor current:

Equivalent circuit diagram during
 $0 < t < dT$



Equivalent circuit diagram during
 $dT < t < T$



Buck-boost - averaging

Derivation $0 < t < dT$

inductor voltage:

$$V_{in}$$

capacitor current:

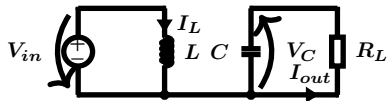
Derivation $dT < t < T$

inductor voltage:

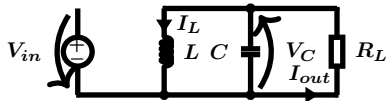
$$-V_C$$

capacitor current:

Equivalent circuit diagram during
 $0 < t < dT$



Equivalent circuit diagram during
 $dT < t < T$



Buck-boost - averaging

Derivation $0 < t < dT$

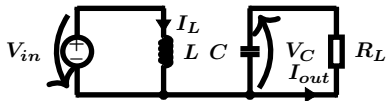
inductor voltage:

$$V_{in}$$

capacitor current:

$$-I_{out}$$

Equivalent circuit diagram during $0 < t < dT$



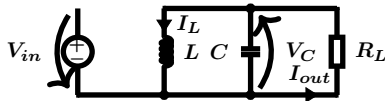
Derivation $dT < t < T$

inductor voltage:

$$-V_C$$

capacitor current:

Equivalent circuit diagram during $dT < t < T$



Buck-boost - averaging

Derivation $0 < t < dT$

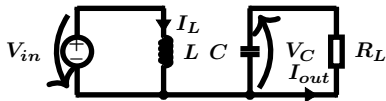
inductor voltage:

$$V_{in}$$

capacitor current:

$$-I_{out}$$

Equivalent circuit diagram during
 $0 < t < dT$



Derivation $dT < t < T$

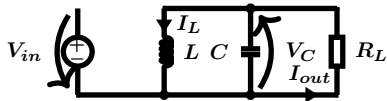
inductor voltage:

$$-V_C$$

capacitor current:

$$I_L - I_{out}$$

Equivalent circuit diagram during
 $dT < t < T$



Buck-boost - averaging

Derivation $0 < t < dT$

inductor voltage:

$$V_{in}$$

capacitor current:

$$-I_{out}$$

Derivation $dT < t < T$

inductor voltage:

$$-V_C$$

capacitor current:

$$I_L - I_{out}$$

Derivation average inductor voltage & capacitor current

$$V_{L_{avg}} = dTV_{in} - (1 - d)TV_C \stackrel{!}{=} 0$$

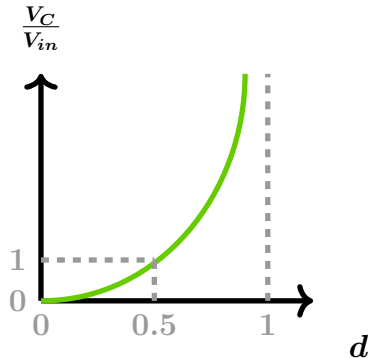
$$I_{C_{avg}} = -dT I_{out} + (1 - d)T(I_L - I_{out}) \stackrel{!}{=} 0$$

Buck-boost - averaging

IMPORTANT DC-transfer function of buckboost

$$\frac{V_C}{V_{in}} = \frac{d}{1-d} \qquad \frac{I_L}{I_{out}} = \frac{1}{1-d} \qquad (56)$$

$$d/(1-d)$$



Buck-boost

Black box approach under assumption: efficiency $\eta = 100\%$

$$I_{in} = \frac{V_{out}}{V_{in}} I_{out} = \frac{d}{1-d} I_{out}$$

IMPORTANT Input current

$$I_{in} = \frac{d}{1-d} I_{out} \quad (57)$$

Inductor current

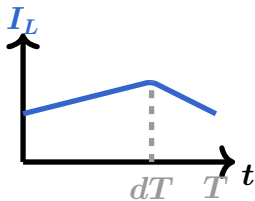
$$I_L = \frac{1}{1-d} I_{out} = \frac{1-d+d}{1-d} I_{out} = \left(1 + \frac{d}{1-d}\right) I_{out}$$

IMPORTANT Inductor current

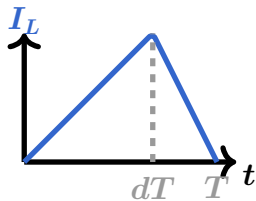
$$I_L = I_{in} + I_{out}$$

Buckboost - all operating modes

Continuous Conduction Mode (CCM)



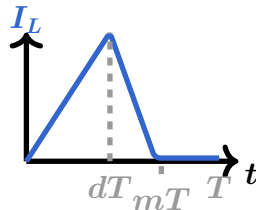
Boundary Conduction Mode (BCM)



also

called Critical Conduction Mode (CrCM) (sometimes abbreviated CCM)

Discontinuous Conduction Mode (DCM)



analysis above not valid here! (later in course!)

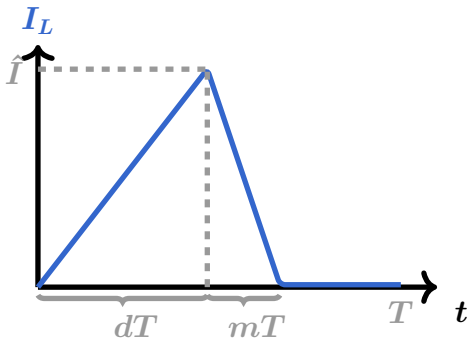
Section 6

Parameters of Converters

- **Nonisolated Converters**
- **Parameters of Converters**
 - Discontinuous Conduction Mode
 - Synchronous Rectification
 - Interleave

Discontinuous Conduction Mode

Inductor current - waveform

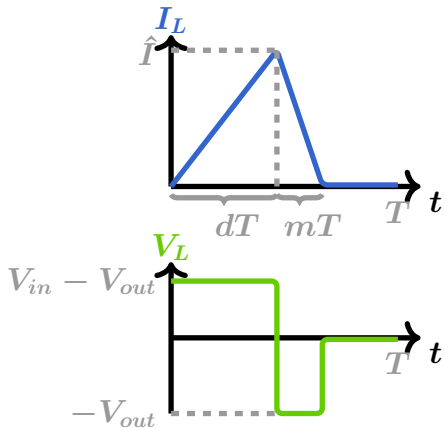


Remaining pre- conditions in DCM

- 1 Ideal power converters are 100 % efficient
 $\Rightarrow P_{in} = P_{out}$
 $\Rightarrow V_{out} = \frac{I_{in}}{I_{out}} V_{in}.$
- 2 Ohms law across the load resistor: $I_{out_{avg}} = \frac{V_{out}}{R_L}.$

Buck in discontinuous conduction mode

Voltage & current across & through inductor



Derivation Input current

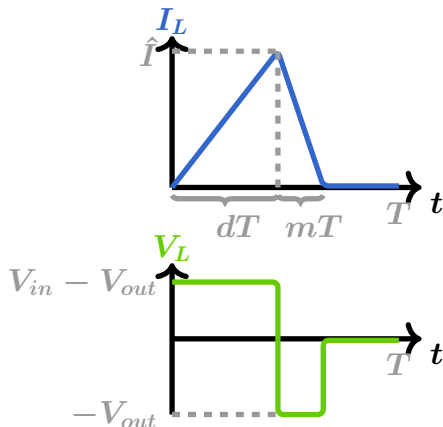
③ peak input current:

$$I_{in_{peak}} = \frac{V_{in} - V_{out}}{L} dT$$

④ mean input current:

Buck in discontinuous conduction mode

Voltage & current across & through inductor



Derivation Input current

③ peak input current:

$$I_{in_{peak}} = \frac{V_{in} - V_{out}}{L} dT$$

④ mean input current:

$$I_{in_{avg}} = \frac{V_{in} - V_{out}}{2L} d^2 T$$

Buck in discontinuous conduction mode

Derivation DC transfer function

Putting 4 and 2 into 1:

$$V_{out} = \frac{V_{in} - V_{out}}{V_{out}} \underbrace{\frac{R}{2L} d^2 T}_{\alpha} V_{in}$$

$$V_{out}^2 + \alpha V_{in} V_{out} - \alpha V_{in}^2 = 0$$

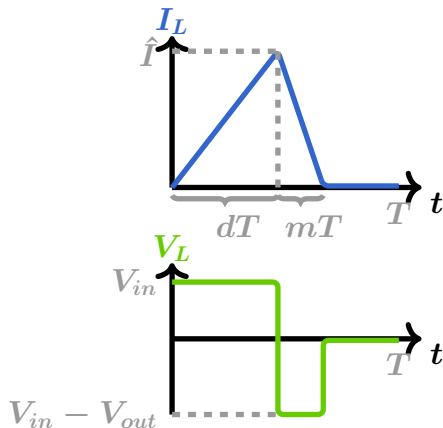
$$V_{out} = \frac{-\alpha V_{in} \pm \sqrt{\alpha^2 V_{in}^2 + 4\alpha V_{in}^2}}{2}$$

IMPORTANT DC transfer function of buck in discontinuous conduction mode

$$V_{out} = \frac{V_{in}}{2} \left(\sqrt{\alpha^2 + 4\alpha} - \alpha \right) \quad (59)$$

Boost in discont. conduction mode

Voltage & current across & through inductor



Derivation Input current

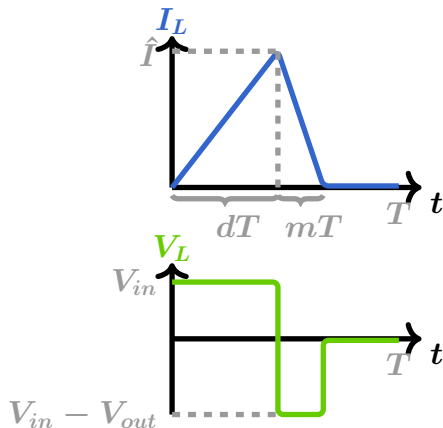
③ peak input current:

$$I_{in_{peak}} = \frac{V_{in}}{L} dT$$

④ mean input current:

Boost in discont. conduction mode

Voltage & current across & through inductor



Derivation Input current

③ peak input current:

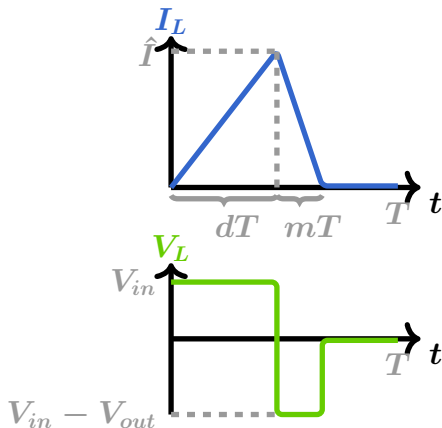
$$I_{in_{peak}} = \frac{V_{in}}{L} dT$$

④ mean input current:

$$I_{in_{avg}} = \frac{V_{in}}{2L} d(d+m)T$$

Boost in discont. conduction mode

Voltage & current across & through inductor



Derivation Input current

③ peak input current:

$$I_{in_{peak}} = \frac{V_{in}}{L} dT$$

④ mean input current:

$$I_{in_{avg}} = \frac{V_{in}}{2L} d(d + m) T$$

compared to buck, one more unknown: m , therefore one relation needed to solve the set of equations

Boost in scont. conduction mode

Derivation Average output current in boost

Average output current is known from 2, but can also be expressed from average diode current:

⑤ average output current:

$$I_{out_{avg}} = \frac{m}{2} I_{in_{peak}} = \frac{V_{in}}{2L} m d T$$

Boost in discontin. conduction mode

Derivation Average output current in boost

Average output current is known from 2, but can also be expressed from average diode current:

⑤ average output current:

$$I_{out_{avg}} = \frac{m}{2} I_{in_{peak}} = \frac{V_{in}}{2L} m dT$$

solving for m and using 2 as the other relation for the average output current gives:

$$m = \frac{2L}{R} \frac{1}{dT} \frac{V_{out}}{V_{in}}$$

Boost in discont. conduction mode

Derivation DC transfer function

Now putting m into 4 and using the result together with 2 in 1 gives:

$$V_{out} = \frac{\frac{V_{in}}{2L} d \left(d + \frac{2L}{R} \frac{1}{dT} \frac{V_{out}}{V_{in}} \right) T}{V_{out}} R V_{in}$$

$$V_{out}^2 = \underbrace{\frac{R}{2L} d^2 T}_{\alpha} V_{in}^2 + V_{in} V_{out}$$

$$V_{out}^2 - V_{in} V_{out} - \alpha V_{in}^2 = 0$$

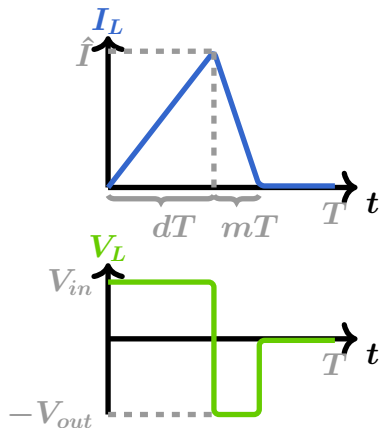
$$V_{out} = \frac{V_{in} \pm \sqrt{V_{in}^2 + 4\alpha V_{in}^2}}{2}$$

IMPORTANT DC transfer function of boost in DCM

$$V_{out} = \frac{V_{in}}{2} \left(1 + \sqrt{1 + 4\alpha} \right) \quad (60)$$

Buck-boost in DCM

Voltage & current across & through inductor



Derivation Input current

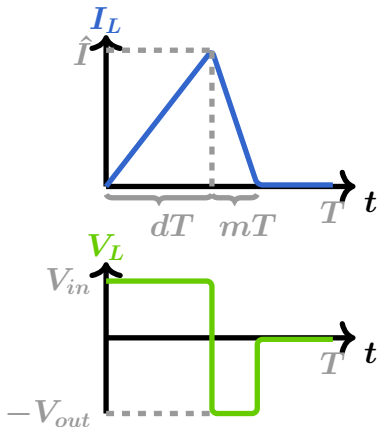
③ peak input current:

$$I_{in_{peak}} = \frac{V_{in}}{L} dT$$

④ mean input current:

Buck-boost in DCM

Voltage & current across & through inductor



Derivation Input current

③ peak input current:

$$I_{in_{peak}} = \frac{V_{in}}{L} dT$$

④ mean input current:

$$I_{in_{avg}} = \frac{V_{in}}{2L} d^2 T$$

Buck-boost in DCM

Derivation DC transfer function

Putting 4 and 2 into 1:

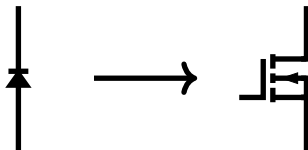
$$V_{out} = \frac{V_{in}^2 R}{2LV_{out}} d^2 T$$

IMPORTANT DC transfer function of buck-boost in discontinuous conduction mode

$$V_{out} = dV_{in} \sqrt{\frac{RT}{2L}} \quad (61)$$

Synchronous rectification

Replacement of diode with MOSFET



Advantages

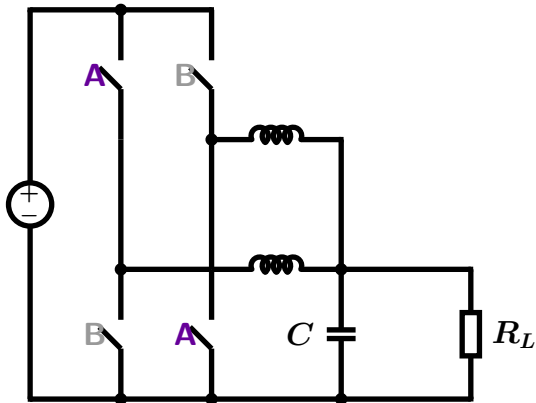
- lower conduction losses
- enables bipolar output, e.g. for motor drives or amplifiers

Disadvantages

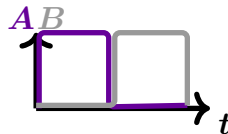
- needs an extra gate driver, i.e. more gate drive losses
- potentially complex control as it is always in continuous conduction mode

Interleave

Interleaved buck

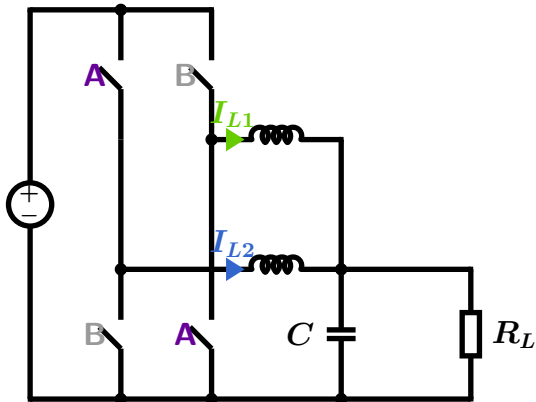


Gate voltages

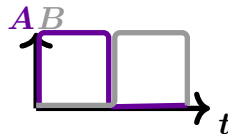


Interleave

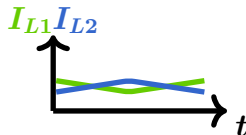
Interleaved buck



Gate voltages



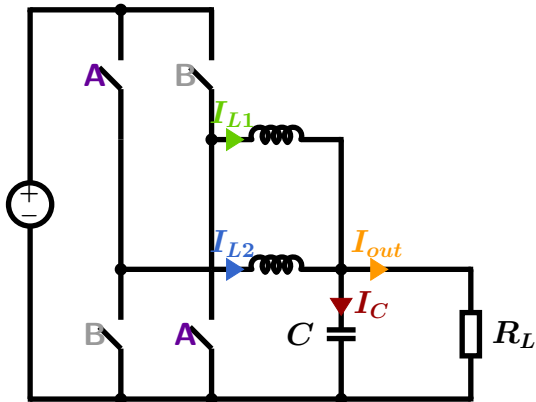
Currents $d = 0.5$



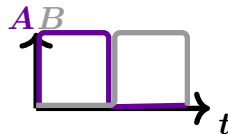
Parameters of Converters

Interleave

Interleaved buck



Gate voltages



Currents $d = 0.5$



Buck in various conduction modes

Exercise 6.1 Buck

An ideal buck converter provides $V_{out} = 5 \text{ V}$ from $V_{in} = 12 \text{ V}$ while it operates with a period of $T = 10 \text{ }\mu\text{s}$. The inductor is $L = 58.1 \text{ }\mu\text{H}$, while the output capacitor is assumed to be infinite.

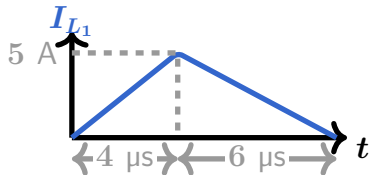
- Plot the inductor current for a load current $I_{out} = 5 \text{ A}$.
- The output current is reduced to the point, where the inductor current hits zero exactly at the end of the period. Draw the inductor current for this constellation.
- How big is the load resistance in this case?
- The load resistance is increased further: now the peak current of the inductor is $\hat{I}_L = 354 \text{ mA}$. How long does the transistor conduct? Draw the inductor current.
- How big is the average inductor current and load resistance now?

Boost

Exercise 6.2 Boost in BCM...

The diagram shows the current waveform of an ideal boost, powered by a constant DC voltage $V_{in} = 180 \text{ V}$ with infinite output capacitance and finite inductance L_1 .

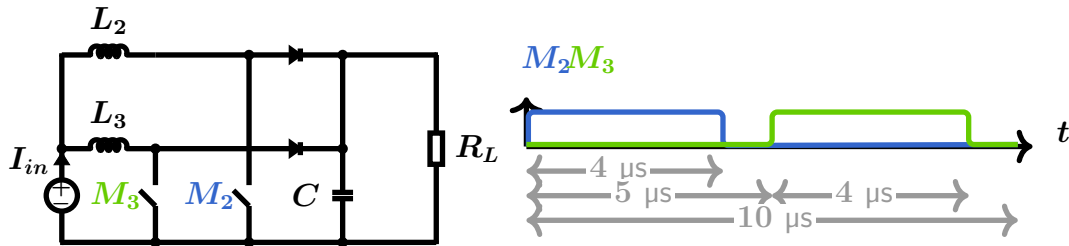
- What is the inductance L_1 ?
- What is the output voltage?
- What is the load resistance?
- Now the output power shall be doubled but the converter shall maintain BCM. Therefore a new inductor L_a is designed using the same core material and size as L_1 . Therefore $L_a = \frac{L_1}{2}$. How does the number of turns change from L_1 to L_a ?



Boost

Exercise 6.2 ...and interleaved boost

The inductance calculated in a) is used for L_2 & L_3 in an interleaved boost:



- e Draw the input current I_{in} of the converter and label its maximum and minimum values.
- f What is the RMS-value of the currents in the MOSFETs?

Section 7

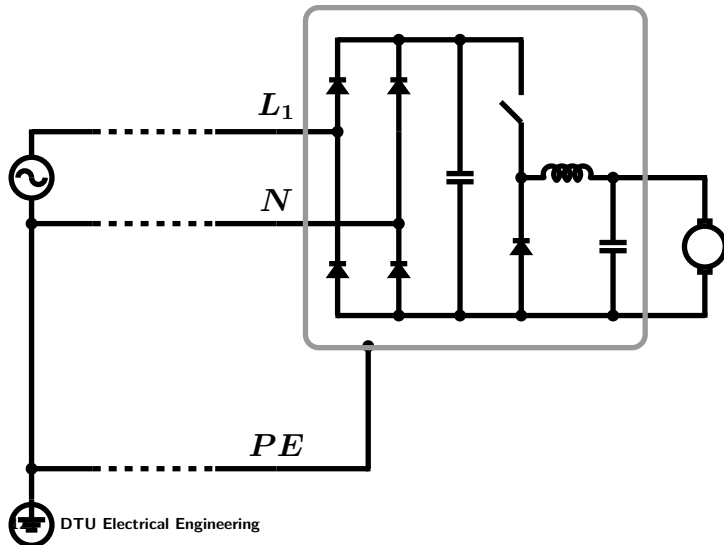
Isolated Converters

- Nonisolated Converters
- Parameters of Converters

Isolated Converters

Isolation

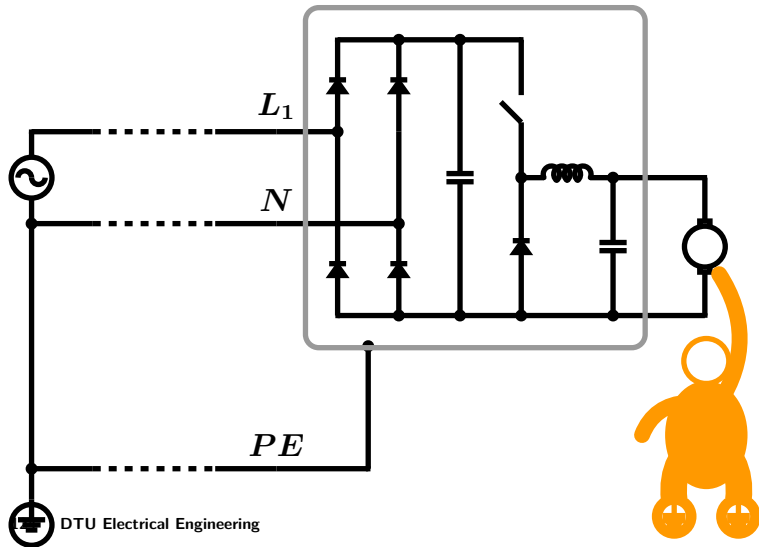
Failure of switch in a buck



Isolated Converters

Isolation

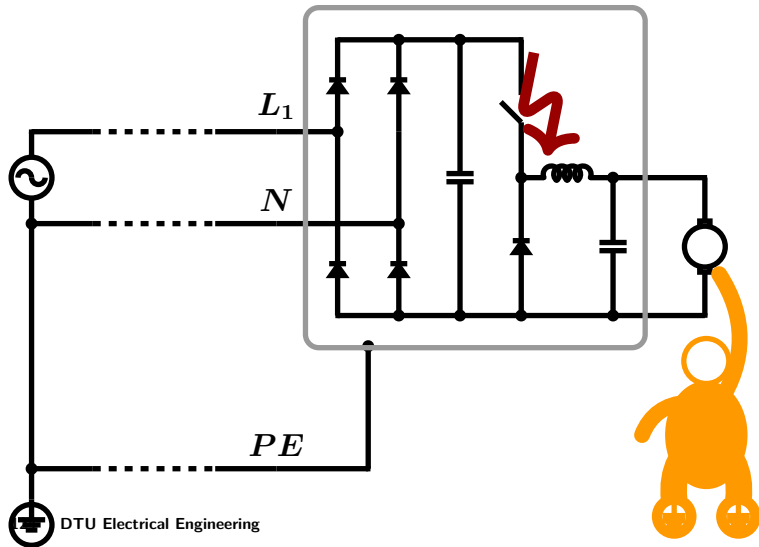
Failure of switch in a buck



Isolated Converters

Isolation

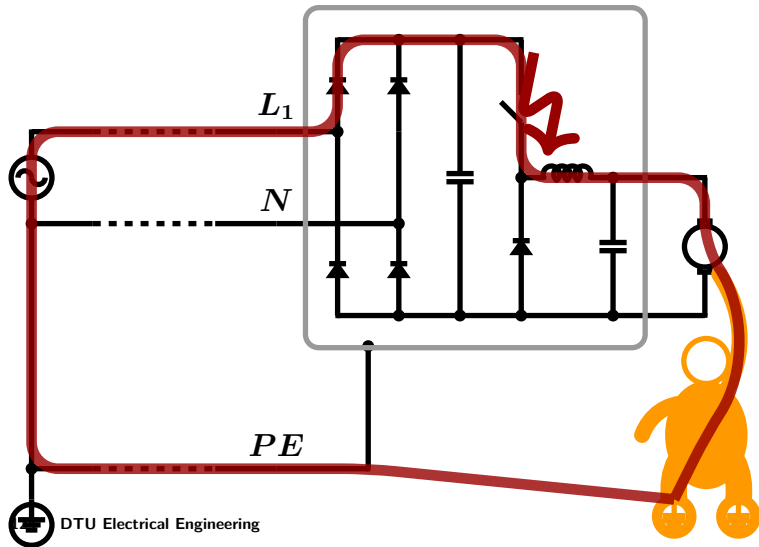
Failure of switch in a buck



Isolated Converters

Isolation

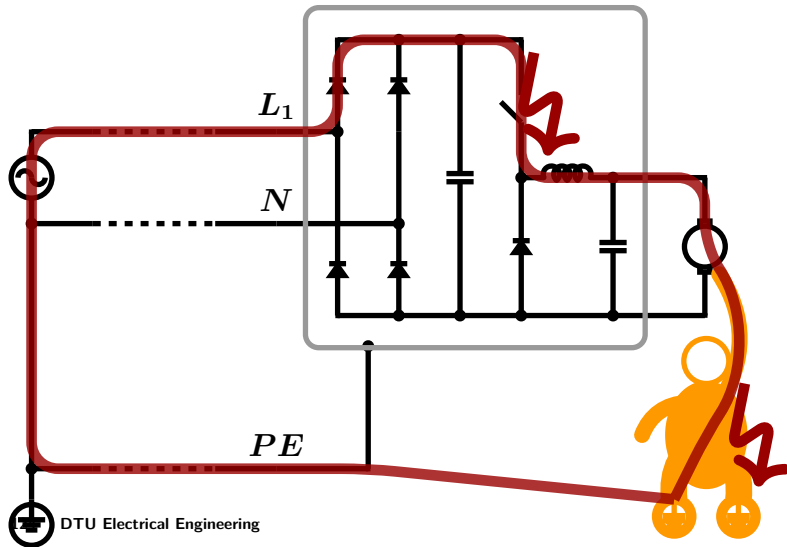
Failure of switch in a buck



Isolated Converters

Isolation

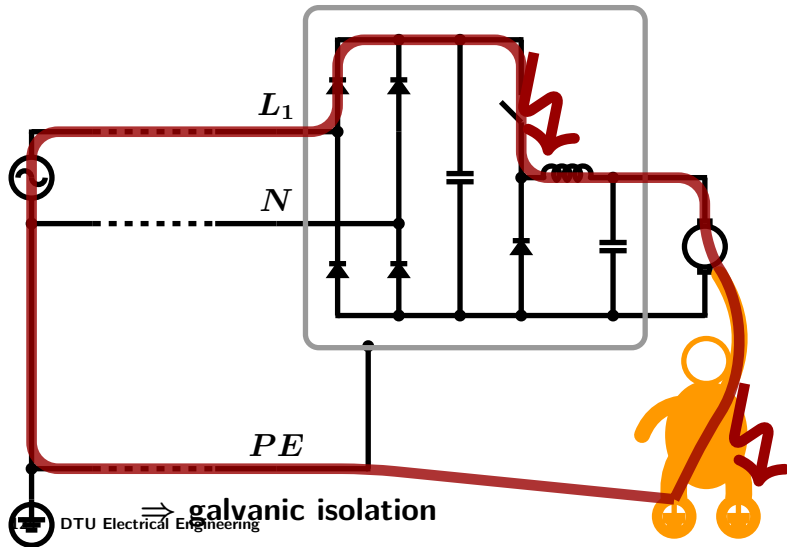
Failure of switch in a buck



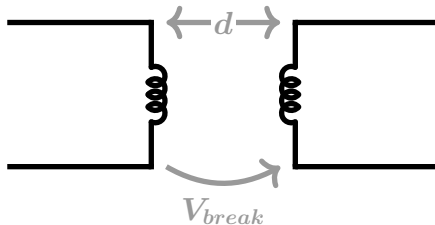
Isolated Converters

Isolation

Failure of switch in a buck



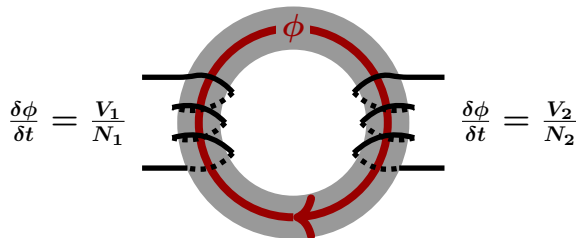
Transformer



- creepage & clearance distance
- minimum breakdown voltage
- no DC energy transfer

energy conversion: electric $\rightarrow$ magnetic $\rightarrow$ electric

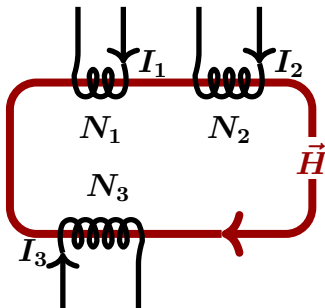
Faradays law



IMPORTANT Constant turns voltage

$$\frac{V_1}{N_1} = \frac{V_2}{N_2} = \dots = \frac{V_n}{N_n} = \frac{\delta\phi}{\delta t}$$

Amperes law



IMPORTANT Sum of all turns-current products is equal to externally provided magnetic flux (or zero if there is no external ϕ)

$$\sum_{n=1}^{\infty} N_n I_n = \vec{H} \delta l = \frac{\phi}{A_L}$$

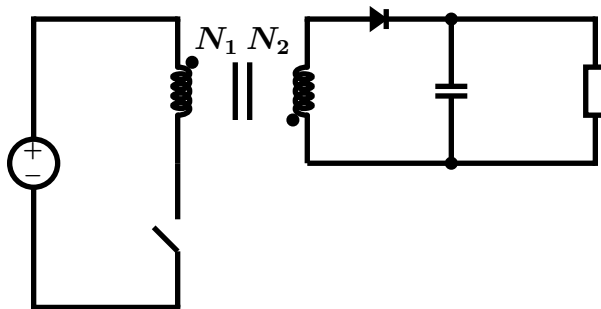
Other means of isolation

- **optocoupler:**
electric $\rightarrow$ optic $\rightarrow$ electric
- **piezotransformer:**
electric $\rightarrow$ mechanic $\rightarrow$ electric

Isolated Converters

Flyback

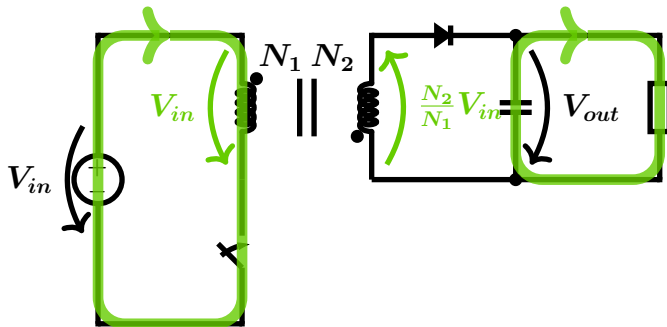
Circuit diagram



Isolated Converters

Flyback

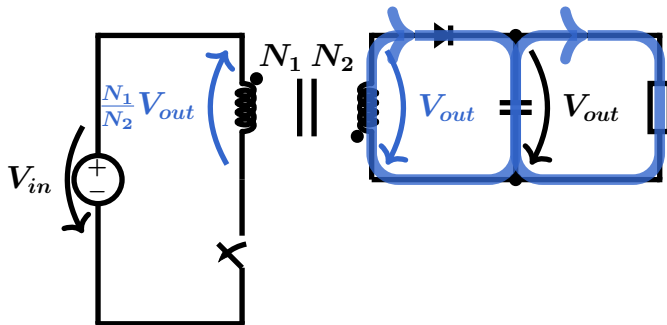
Operation $0 < t < dT$



Isolated Converters

Flyback

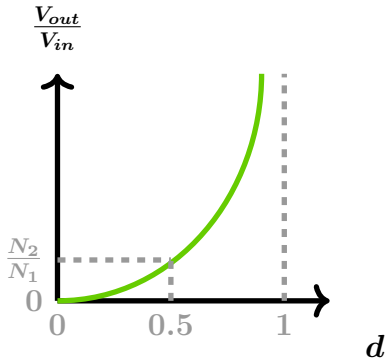
Operation $dT < t < T$



IMPORTANT DC transfer function

$$\frac{V_{out}}{V_{in}} = \frac{N_2}{N_1} \frac{d}{1-d}$$

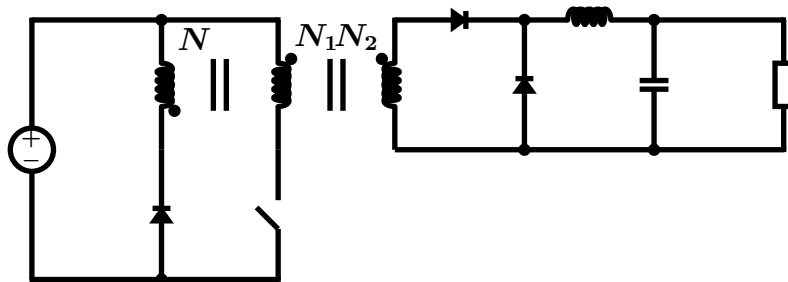
$d/(1-d)$



Isolated Converters

Forward

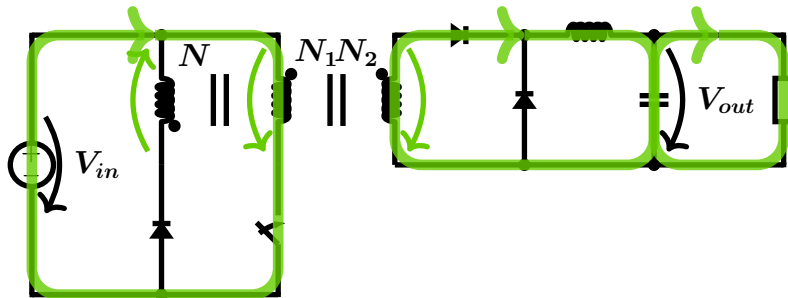
Circuit diagram



Isolated Converters

Forward

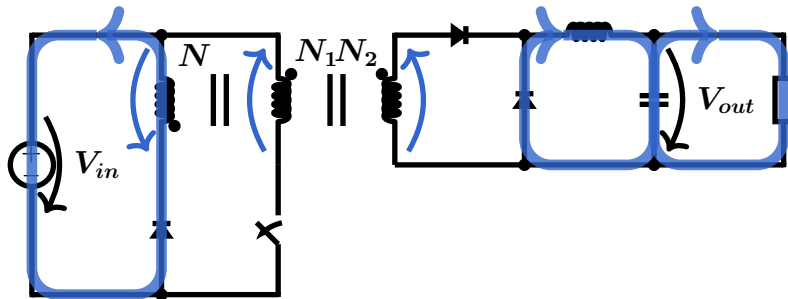
Operation $0 < t < dT$



Isolated Converters

Forward

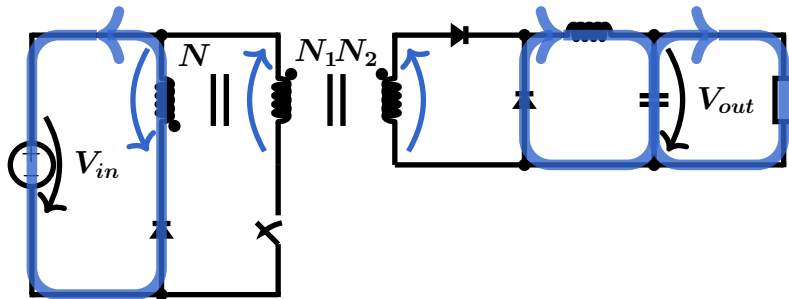
Operation $dT < t < T$



Isolated Converters

Forward

Operation $dT < t < T$



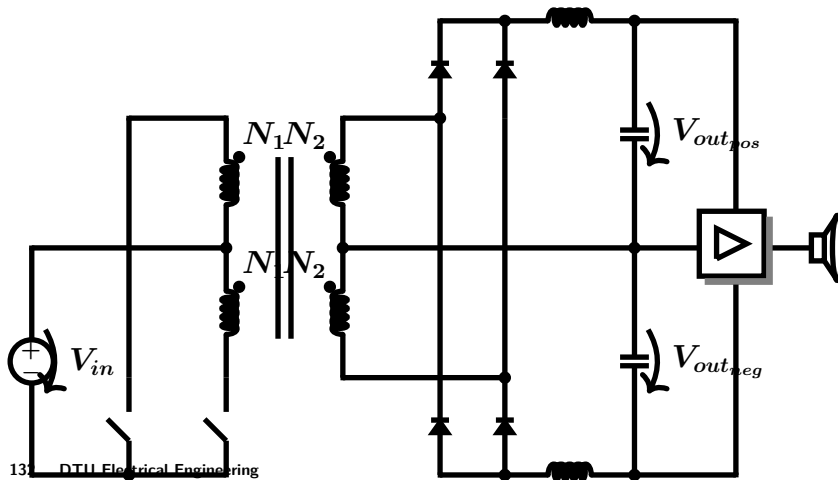
IMPORTANT DC transfer function

$$\frac{V_{out}}{V_{in}} = \frac{N_2}{N_1} d$$

where d is limited!

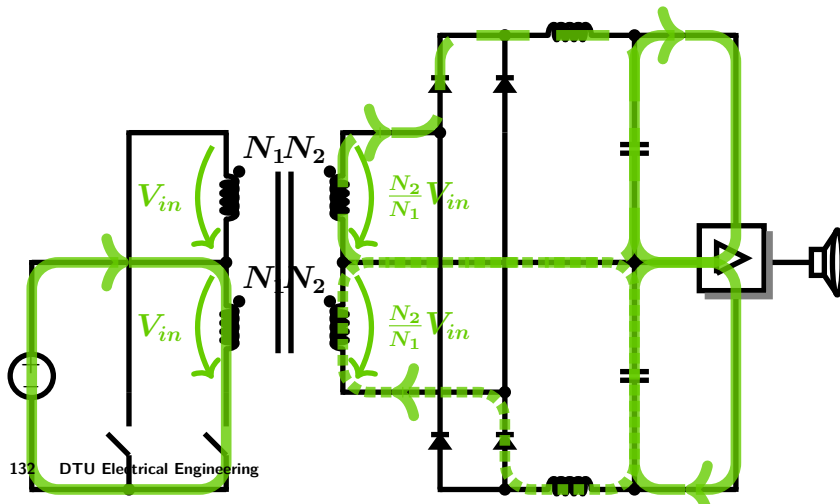
Push-pull with full bridge rectifier and an audio amplifier as load

Circuit diagram



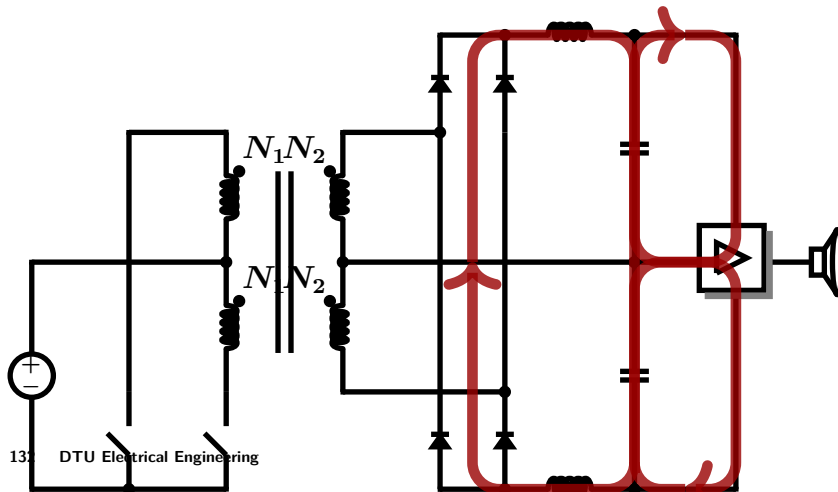
Push-pull with full bridge rectifier and an audio amplifier as load

Operation $0 < t < dT$



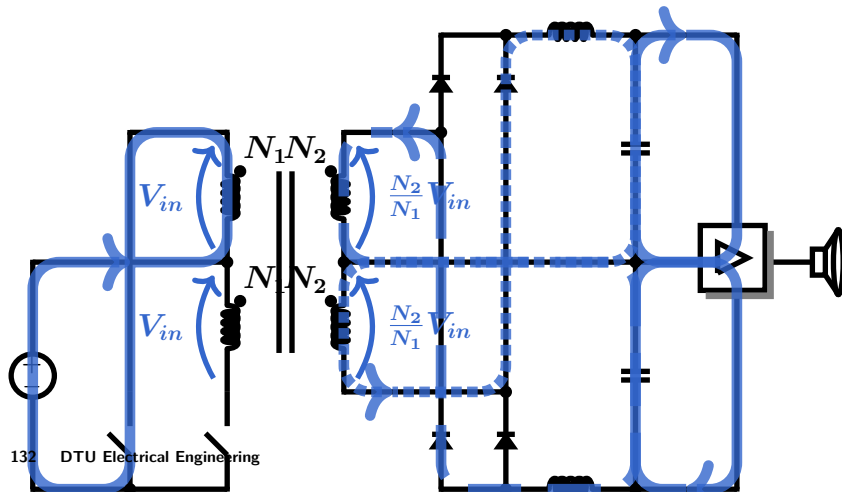
Push-pull with full bridge rectifier and an audio amplifier as load

Operation $dT < t < \frac{T}{2}$



Push-pull with full bridge rectifier and an audio amplifier as load

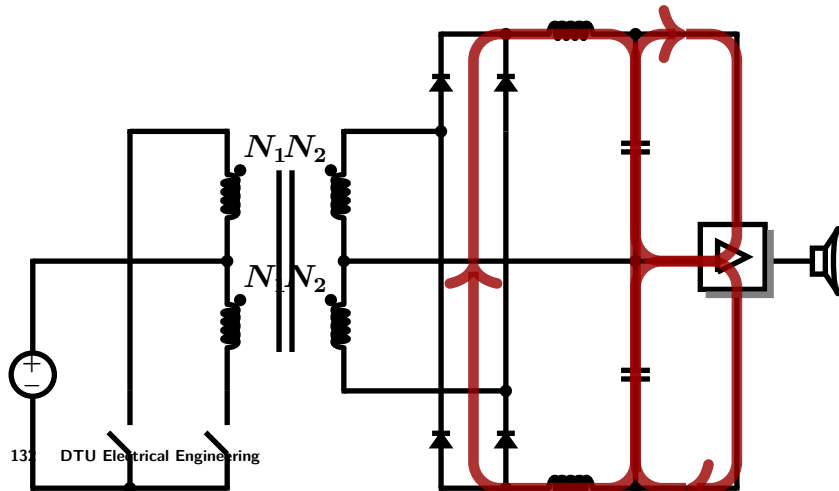
Operation $\frac{T}{2} < t < \frac{T}{2} + dT$



Isolated Converters

Push-pull with full bridge rectifier and an audio amplifier as load

Operation $\frac{T}{2} + dT < t < T$



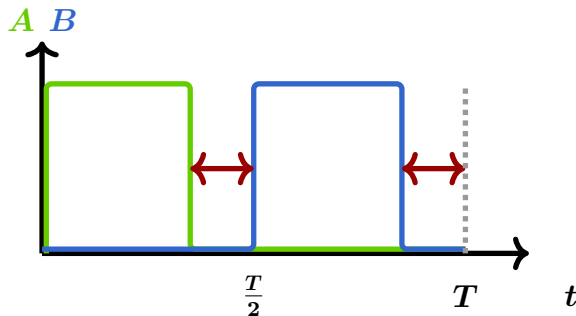
Push-pull

IMPORTANT DC transfer function

$$\frac{V_{out}}{V_{in}} = \frac{N_2}{N_1} d$$

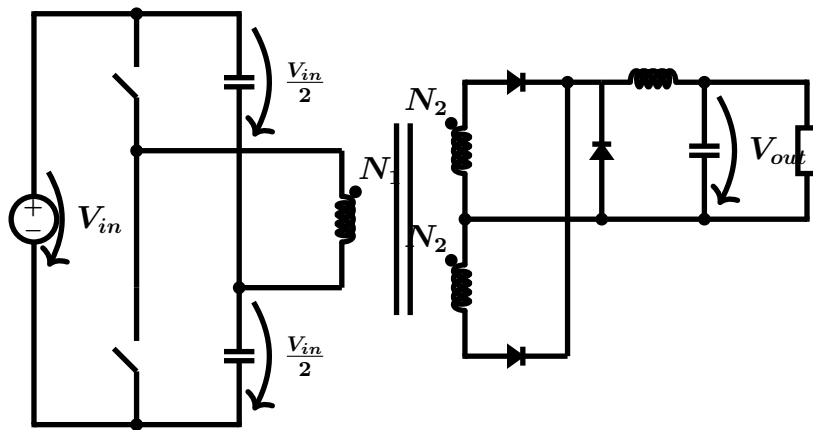
where $d < 0.5$.

Gate signals



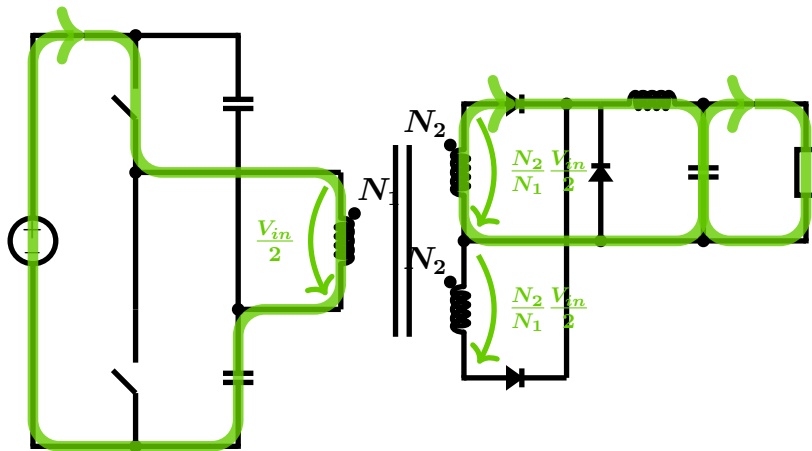
Half bridge with current doubler

Circuit diagram



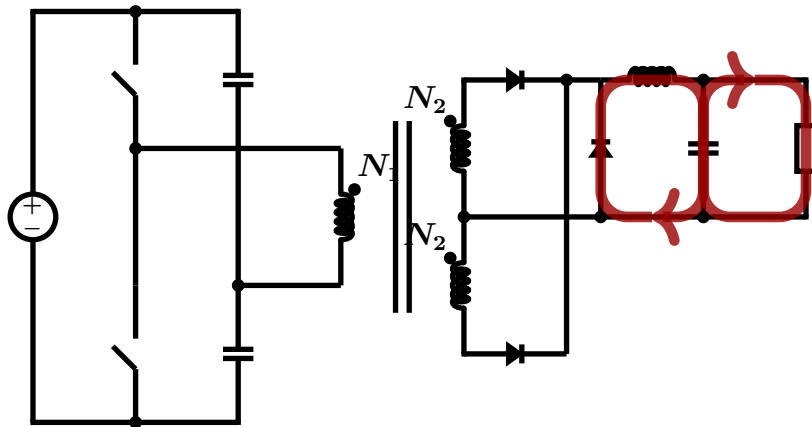
Half bridge with current doubler

Operation $0 < t < dT$



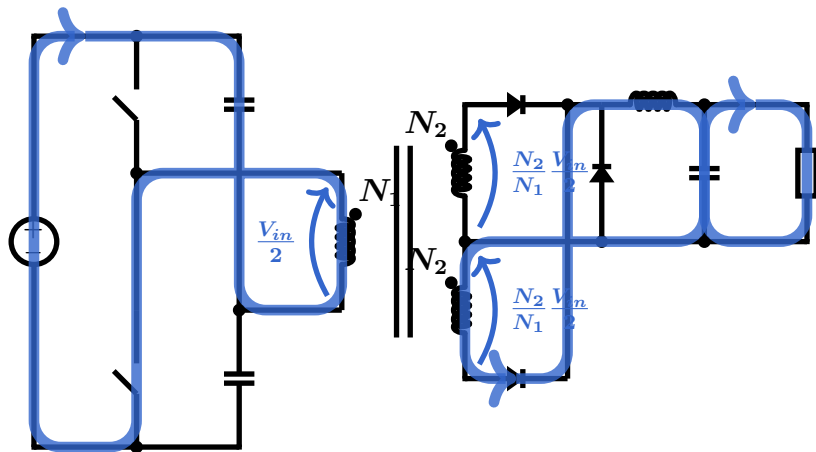
Half bridge with current doubler

Operation $dT < t < \frac{T}{2}$



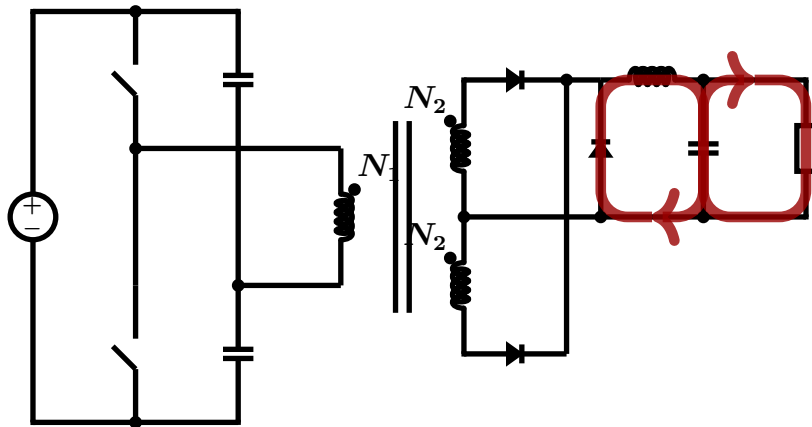
Half bridge with current doubler

Operation $\frac{T}{2} < t < \frac{T}{2} + dT$



Half bridge with current doubler

Operation $\frac{T}{2} + dT < t < T$



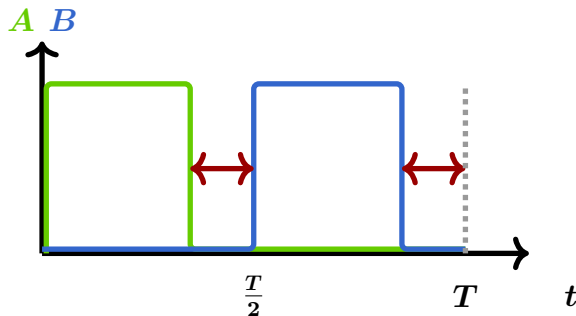
Half bridge

IMPORTANT DC transfer function

$$\frac{V_{out}}{V_{in}} = \frac{N_2}{N_1} d$$

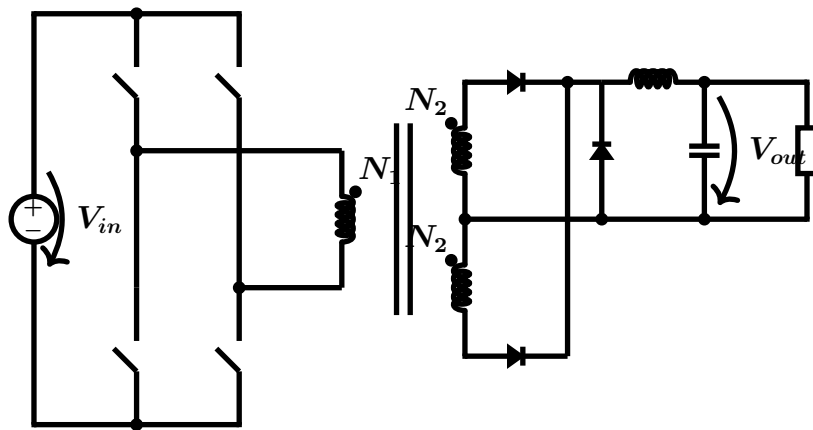
where $d < 0.5$.

Gate signals

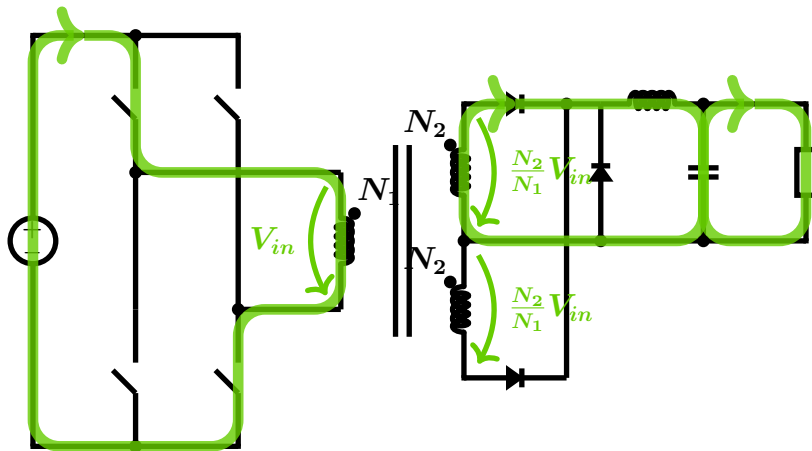


Full bridge with current doubler

Circuit diagram

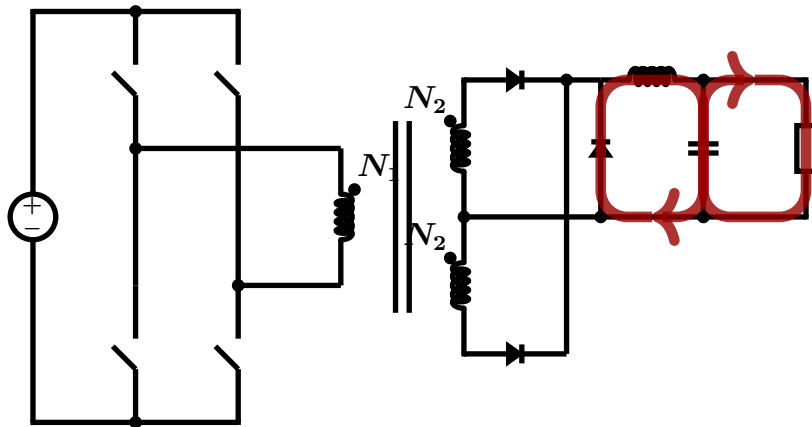


Full bridge with current doubler

Operation $0 < t < dT$ 

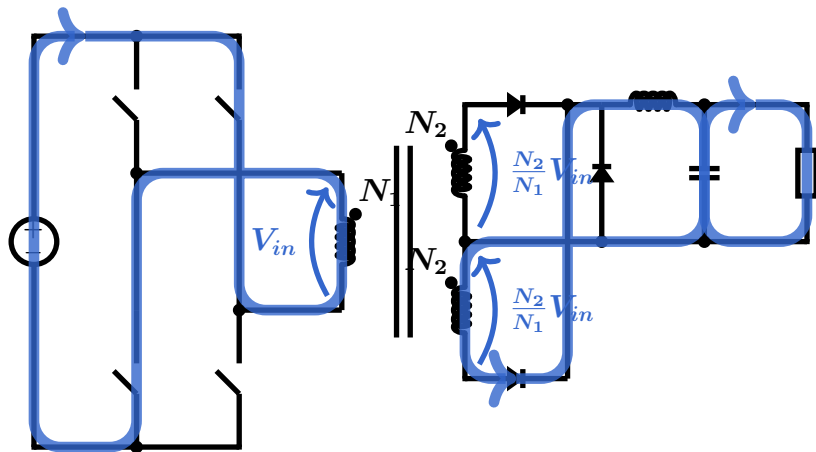
Full bridge with current doubler

Operation $dT < t < \frac{T}{2}$



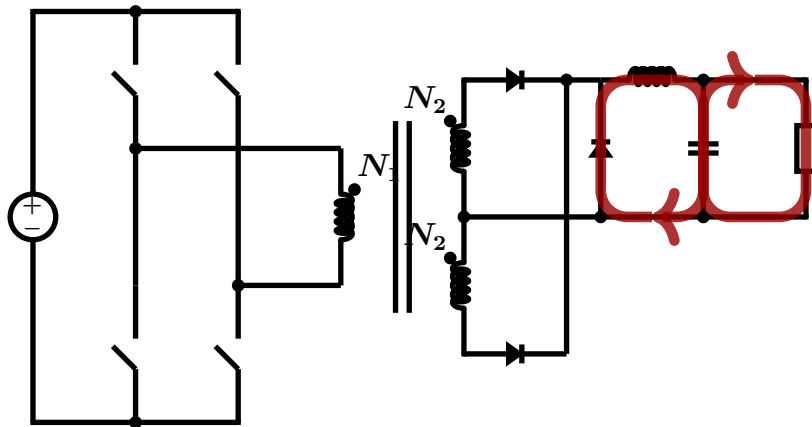
Full bridge with current doubler

Operation $\frac{T}{2} < t < \frac{T}{2} + dT$



Full bridge with current doubler

Operation $\frac{T}{2} + dT < t < T$



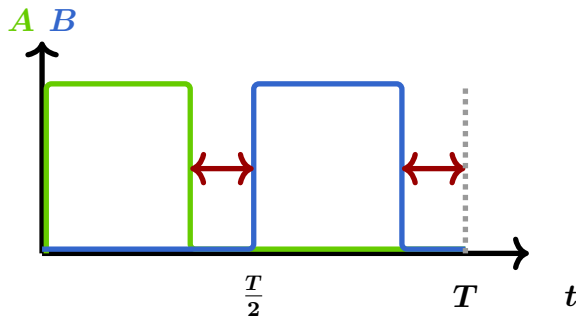
Full bridge

IMPORTANT DC transfer function

$$\frac{V_{out}}{V_{in}} = 2 \frac{N_2}{N_1} d$$

where $d < 0.5$.

Gate signals

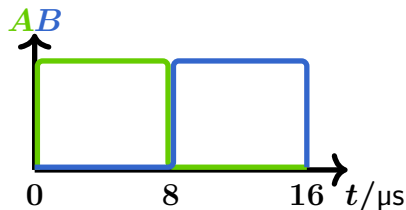
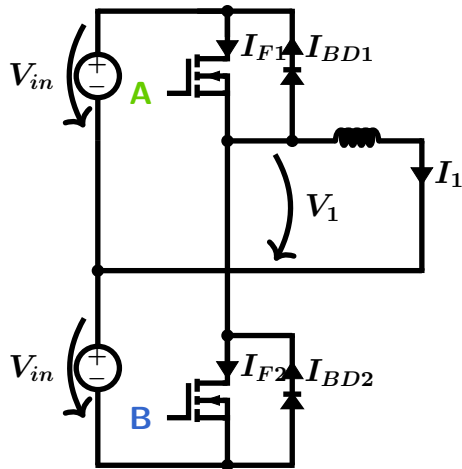


Isolated Converters

Transformer

Exercise 7.1 Transformer

The circuit below consists of ideal components. $V_{in} = 30\text{ V}$ and $L = 12\text{ }\mu\text{H}$.



Isolated Converters

Transformer

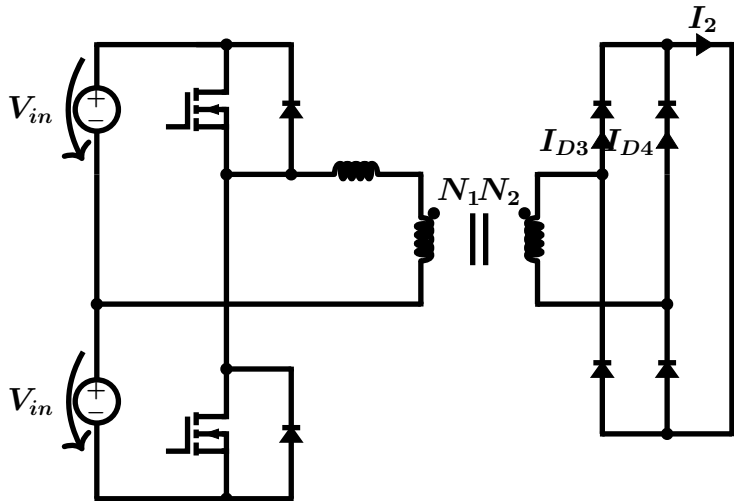
Exercise 7.1 Transformer

- Draw V_1 and I_1 as a function of time, where I_1 is a pure AC current.
- The inductor is dissipating 3 W. Assuming that I_1 does not change, how big is the inductors series resistance to model those losses?
- Now an transformer (without magnetizing inductance) and a shorted full bridge rectifier is added to the circuit, as in the next circuit diagram visualizes. $N_1 = 1$, $N_2 = 2$ and all components are ideal. Draw I_{D3} , I_{D4} and I_2 .

Isolated Converters

Transformer

Exercise 7.1 Transformer



Exercise 7.1 Transformer

- d** Now a 20 V DC source is connected to the diode bridge instead of the short (see next figure). How big is V_L for the four cases:

$$V_3 > 0, I_3 < 0$$

$$V_3 > 0, I_3 > 0$$

$$V_3 < 0, I_3 > 0$$

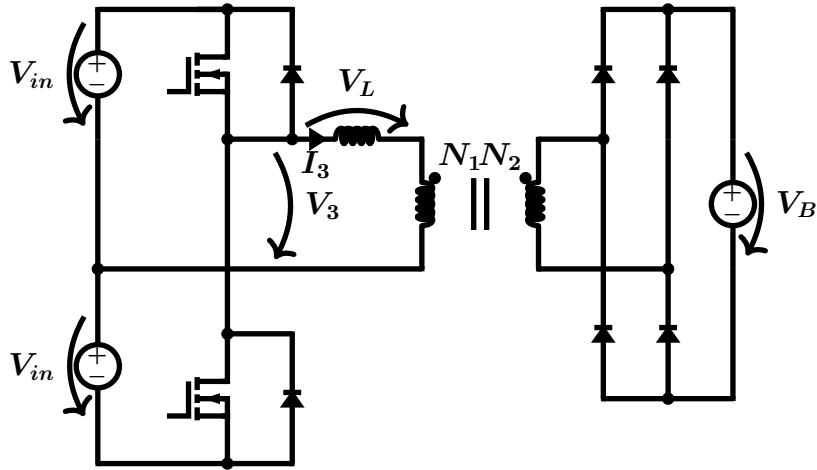
$$V_3 < 0, I_3 < 0$$

- e** Now the inductor current shall peak at ± 10 A. To achieve this, the operation frequency of the transistors changes. At the time, when $I_3 = -10$ A, the voltage V_3 changes from a negative to positive value. Derive the new switching frequency and draw I_3 .

Isolated Converters

Transformer

Exercise 7.1 Transformer



Flyback

Exercise 7.2 Flyback

An ideal flyback operates in discontinuous conduction mode. The output capacitance is assumed infinite, therefore the output voltage is DC only. The transformer has an inductance $L_1 = 1.3 \text{ mH}$ seen from the primary side and $L_2 = 52 \text{ } \mu\text{H}$ seen from the secondary side. The input voltage $V_{in} = 325 \text{ V}$ and the output voltage $V_{out} = 50 \text{ V}$. The transistor conducts $10 \text{ } \mu\text{s}$ and the operating period is $25 \text{ } \mu\text{s}$.

- a** What is the peak input current? Draw the input current.
- b** What is the average input current and how much power does the source provide?
- c** Draw the diode current.

Exercise 7.2 Flyback

- d** How big is the output current?
- e** Draw the voltage across the primary side of the transformer and the diode voltage.
- f** What is the RMS value of the diode current?
- g** What is the RMS value of the capacitor current?

Part IV

Supportive Circuits

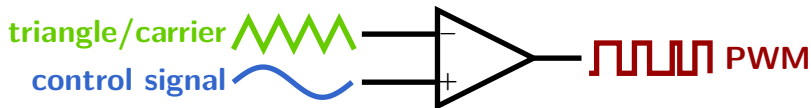
Section 8

Supplementary Circuits

- **Supplementary Circuits**
 - Pulse Width Modulation Generators
 - Gate Drivers
 - Level Shifters
 - Controllers

Pulse width generators

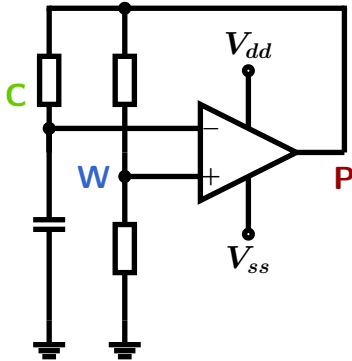
Comparator



- Triangle (or sawtooth) typically provided by RC-oscillator or synchronized to a crystal
- Pulse width modulation (PWM) used to compensate for line-/load-variations and/or AC in- or output.

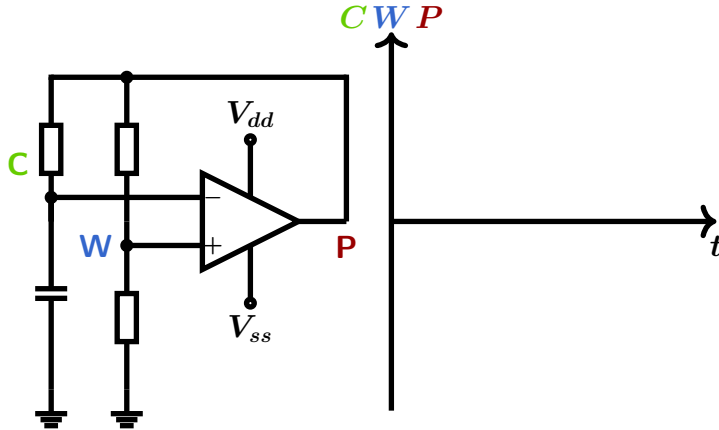
Pulse width generators

Hysteretic self-oscillating modulator



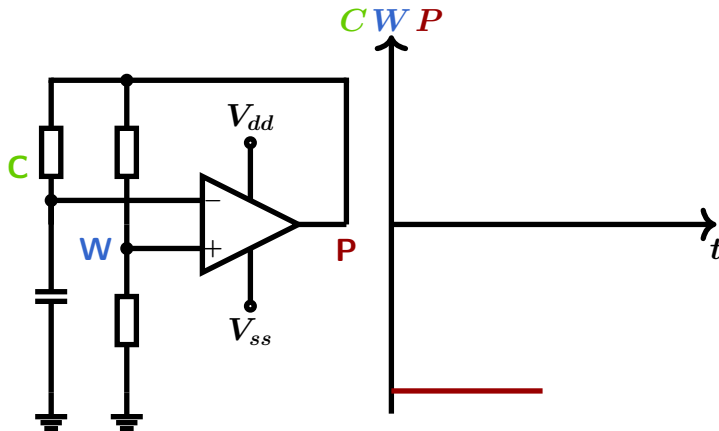
Pulse width generators

Hysteretic self-oscillating modulator



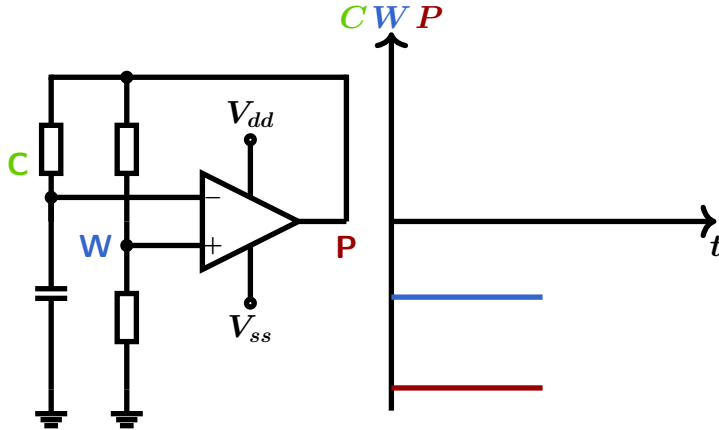
Pulse width generators

Hysteretic self-oscillating modulator



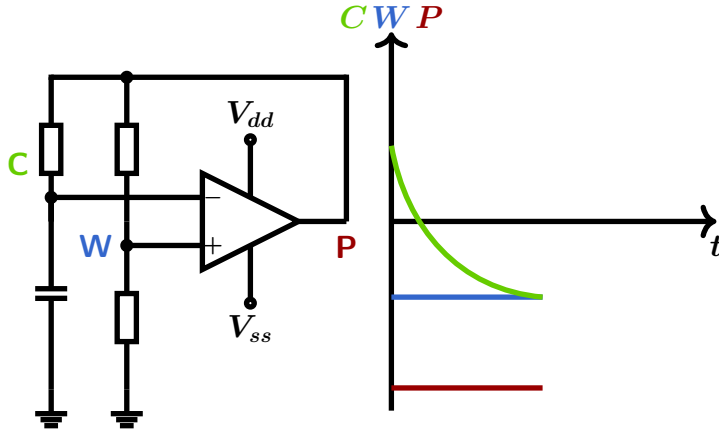
Pulse width generators

Hysteretic self-oscillating modulator



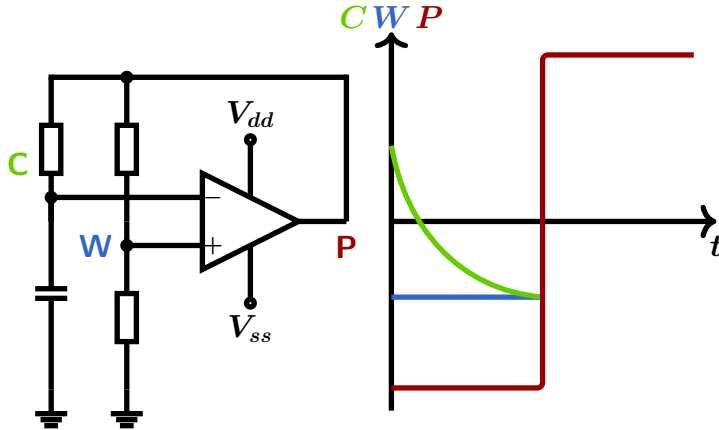
Pulse width generators

Hysteretic self-oscillating modulator



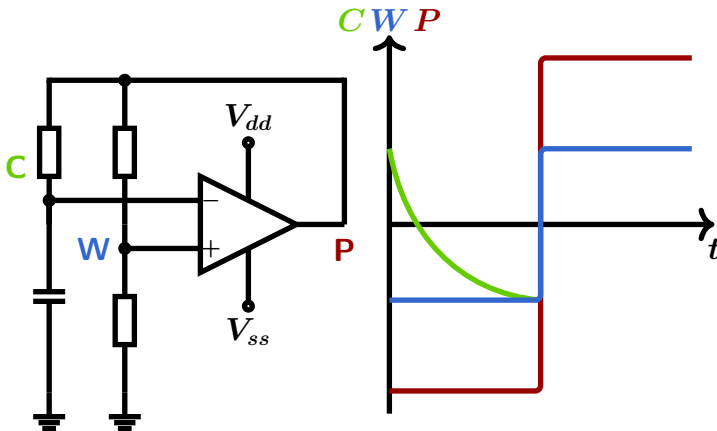
Pulse width generators

Hysteretic self-oscillating modulator



Pulse width generators

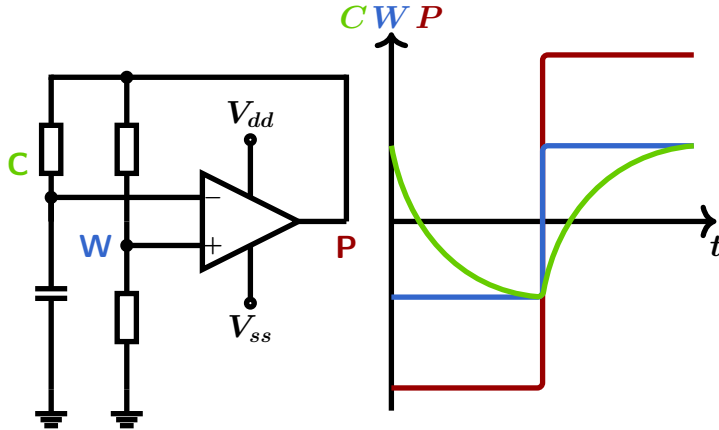
Hysteretic self-oscillating modulator



Supplementary Circuits

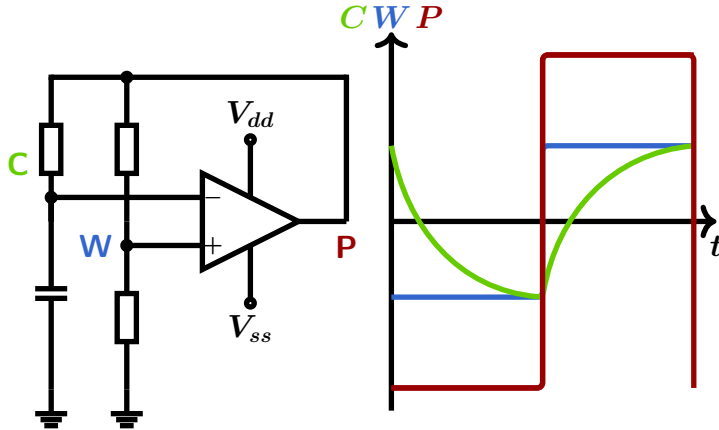
Pulse width generators

Hysteretic self-oscillating modulator



Pulse width generators

Hysteretic self-oscillating modulator

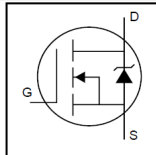


Supplementary Circuits

Gate drivers

Gate drivers

- Advanced Process Technology
- Ultra Low On-Resistance
- Dynamic dv/dt Rating
- 175°C Operating Temperature
- Fast Switching
- Fully Avalanche Rated



$$V_{DS} = 100V$$

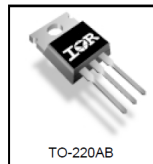
$$R_{DS(on)} = 44m\Omega$$

$$I_D = 33A$$

Description

Advanced HEXFET® Power MOSFETs from International Rectifier utilize advanced processing techniques to achieve extremely low on-resistance per silicon area. This benefit, combined with the fast switching speed and ruggedized device design that HEXFET power MOSFETs are well known for, provides the designer with an extremely efficient and reliable device for use in a wide variety of applications.

The TO-220 package is universally preferred for all commercial-industrial applications at power dissipation levels to approximately 50 watts. The low thermal resistance and low package cost of the TO-220 contribute to its wide acceptance throughout the industry.

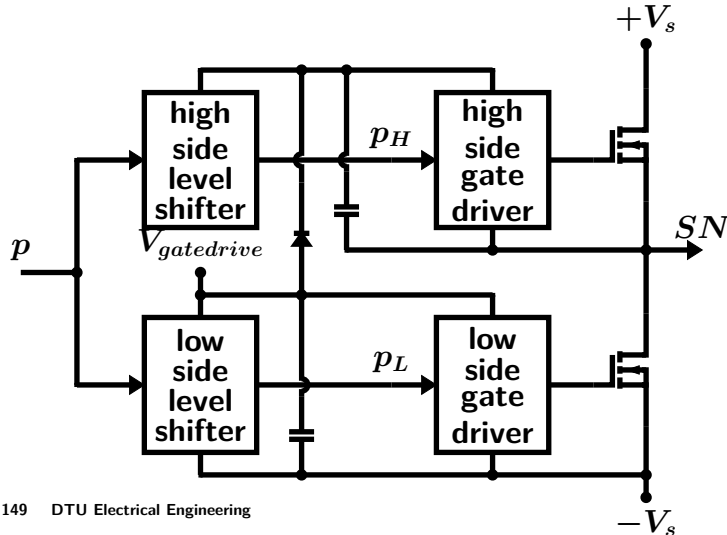


Absolute Maximum Ratings

	Parameter	Max.	Units
$I_D @ T_C = 25^\circ C$	Continuous Drain Current, $V_{GS} @ 10V$	33	A
$I_D @ T_C = 100^\circ C$	Continuous Drain Current, $V_{GS} @ 10V$	23	
I_{DM}	Pulsed Drain Current ①	110	
$P_D @ T_C = 25^\circ C$	Power Dissipation	130	W
	Linear Derating Factor	0.87	W/°C
V_{GS}	Gate-to-Source Voltage	± 20	V

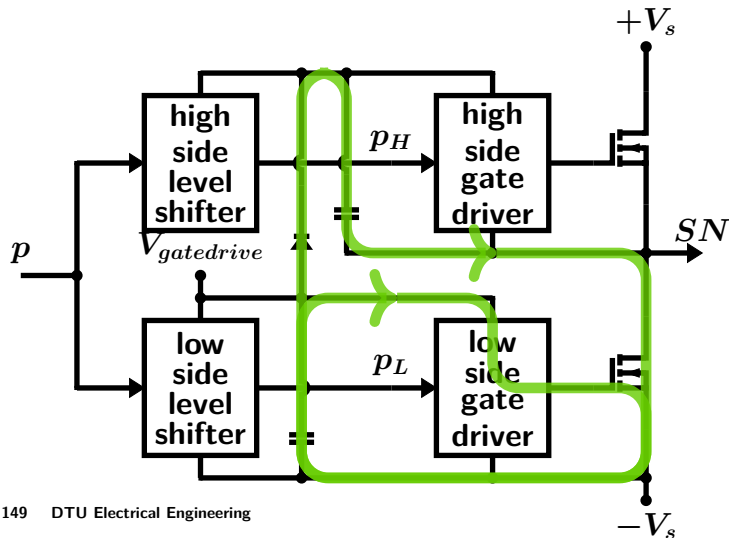
Level shifters

Boot strap in a half bridge



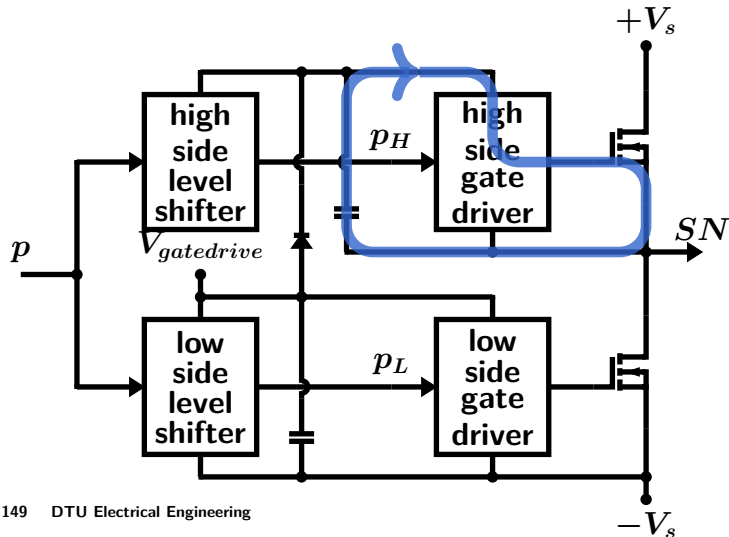
Level shifters

Boot strap in a half bridge

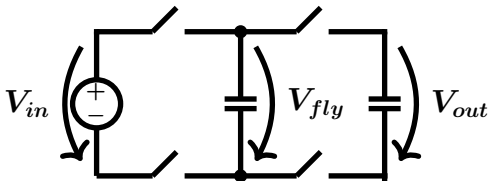


Level shifters

Boot strap in a half bridge

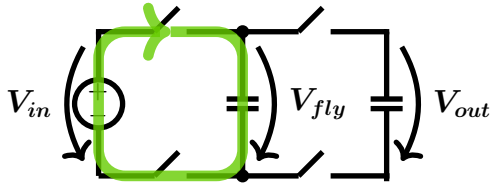


Charge pump



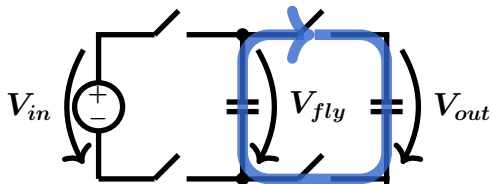
- here a 1:1 converter
- also called switched caps

Charge pump



- here a 1:1 converter
- also called switched caps

Charge pump

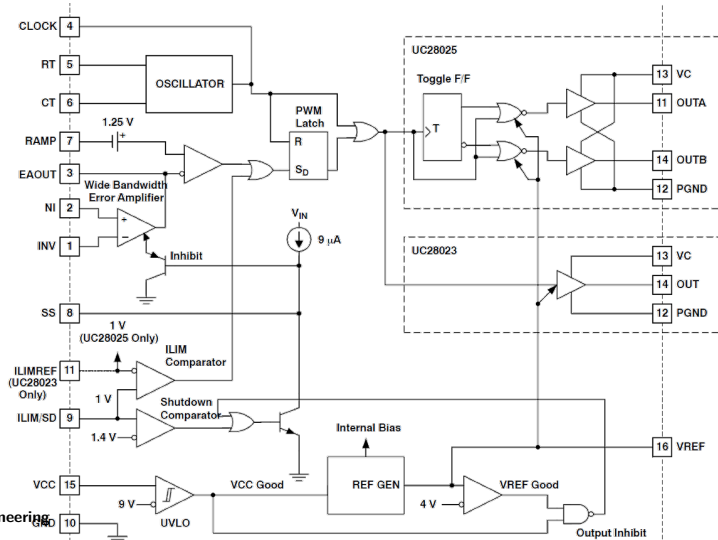


- here a 1:1 converter
- also called switched caps

Supplementary Circuits

Control circuits

Everything in one integrated circuit (IC), e.g. UCC28025



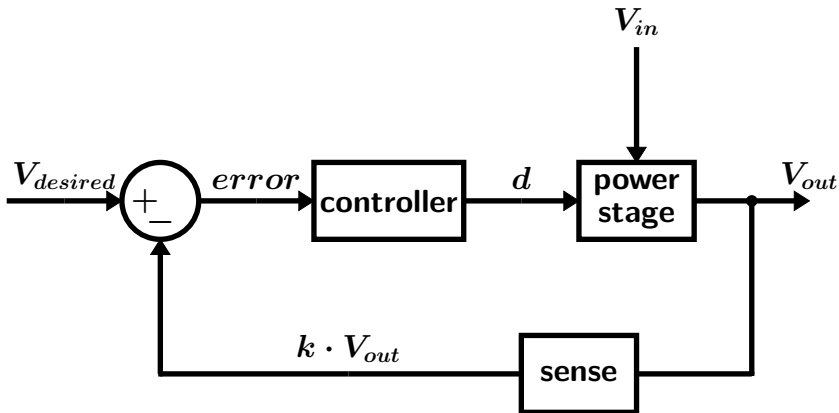
Section 9

Control of Power Converters

- **Supplementary Circuits**

Modelling of the control loop

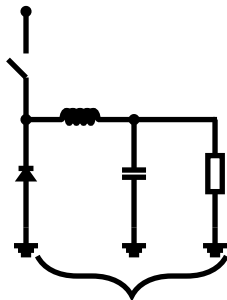
Control loop



Modelling of the control loop

Buck converter

- DC transfer function: $\frac{V_{out}}{V_{in}} = d$
- AC transfer function:



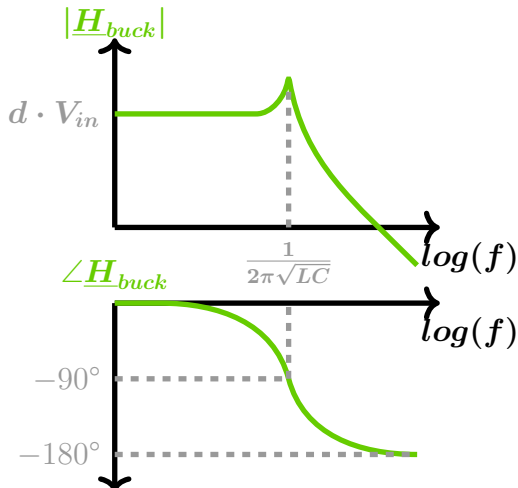
LCR-lowpass $\underline{H}_{LCR} = \frac{1}{1 + s\frac{L}{R} + s^2 LC}$

- combined transfer function: $\underline{H}_{buck} = \frac{d}{1 + s\frac{L}{R} + s^2 LC}$

Modelling the control loop

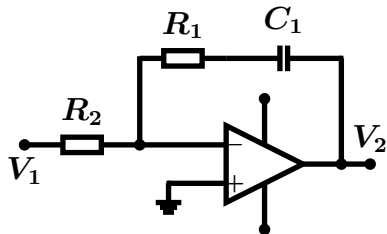
Buck transfer function

$$\underline{H}_{buck} = \frac{d}{1 + s\frac{L}{R} + s^2 LC}$$



PI controller

Implementation



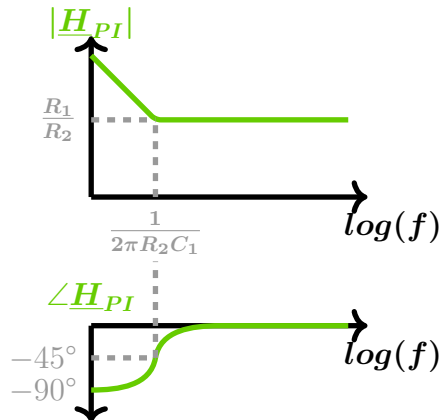
analytically

$$\underline{H}_{PI} = \underbrace{\frac{R_1}{R_2}}_P + \underbrace{\frac{1}{sR_2C_1}}_I$$

P: proportional part

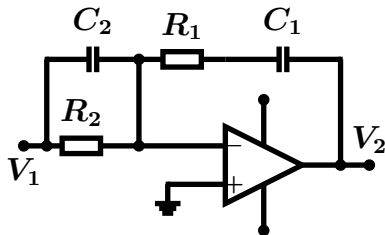
I: integrational part

Transfer Function



PID controller

Implementation



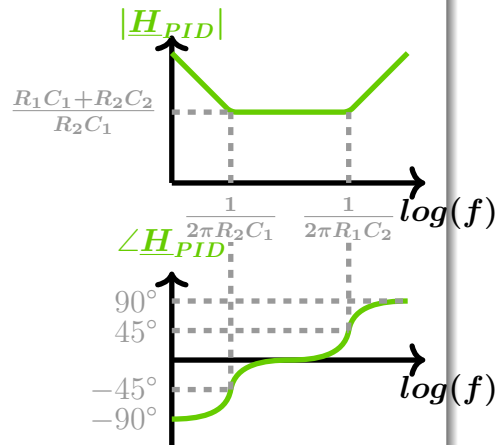
Transfer function analytically

$$\underline{H}_{PI} = \underbrace{\frac{R_1 C_1 + R_2 C_2}{R_2 C_1}}_P + \underbrace{\frac{1}{s R_2 C_1}}_I + \underbrace{\frac{s R_1 C_2}{D}}_D$$

P: proportional, I: integrational part,

D: derivative part

Transfer Function graphically



Resulting loop transfer functions

Exercise 9.1 combining a buck converter with a PI controller

Sketch the amplitude and the phase of the open loop transfer function of a buck $\underline{H}_{buck}$, that is controlled by a PI controller $\underline{H}_{PI}$.

Exercise 9.2 combining a buck converter with a PID controller

Sketch the amplitude and the phase of the open loop transfer function of a buck $\underline{H}_{buck}$, that is controlled by a PID controller $\underline{H}_{PID}$.

Part V

References

Literature

IMPORTANT Primary course literature

The primary literature for the course is [1].

Further reading

Another good book is [2]. It's about the same level as [1].

A more advanced book is [3]. It gets frequently called "the bible" among power electronics experts.

For assistance with LTSpice, [4] is a good reference.

Section 10

Course Literature

- **Course Literature**

Course Literature

References I



Ned Mohan.

Power electronics: a first course.

Wiley, 2011.



Muhammad H Rashid.

Power electronics handbook.

Butterworth-Heinemann, 2017.



Robert W Erickson and Dragan Maksimovic.

Fundamentals of power electronics.

Springer Science & Business Media, 2007.



Erik Bruun.

CMOS Integrated Circuit Simulation with LTspice.

bookboon, 2017.

So long, and Thanks for All the Fish